Project Repository

100 Real-World Projects | GenAI · ML/AI · Data Engineering

Sales & Delivery Intelligence Platform

100	25+	80+
Projects	Domains	Technologies

P001

Customer Churn Prediction Engine

ML/AI

PROBLEM STATEMENT

High customer attrition rate (18% annually) with no early-warning system, resulting in $12M annual revenue loss.

SOLUTION APPROACH

Built an ensemble ML model (XGBoost + LightGBM) trained on 3 years of usage data with real-time scoring via REST API. Deployed on AWS SageMaker with automated retraining pipeline.

PROJECT DETAILS

Domain	Telecom
Team Size	4 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · XGBoost · LightGBM · AWS SageMaker · Apache Kafka · PostgreSQL

EXPECTED OUTCOMES

Reduced churn by 23%, saving ~$2.8M annually. Model achieves 87% recall on at-risk customers.

P001 › Customer Churn Prediction Engine — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- XGBoost
- LightGBM
- AWS SageMaker
- Apache Kafka
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Reduced churn by 23%, saving ~$2.8M annually. Model achieves 87% recall on at-risk customers.

4	3	Telecom
Team Size	Duration	Domain

PROBLEM CONTEXT

High customer attrition rate (18% annually) with no early-warning system, resulting in $12M annual revenue loss.

P002

NLP-Powered Contract Review System

NLP / GenAI

PROBLEM STATEMENT

Legal team spending 2,000+ hours/month manually reviewing vendor contracts for compliance clauses and risk flags.

SOLUTION APPROACH

Fine-tuned BERT model for clause classification and risk scoring. Integrated with SharePoint via Power Automate. Built React dashboard for review workflow.

PROJECT DETAILS

Domain	Legal / FinTech
Team Size	3 people
Duration	2 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · BERT · HuggingFace Transformers · FastAPI · React · SharePoint

EXPECTED OUTCOMES

90% reduction in manual review time. Processes 500+ contracts/day with 94% clause classification accuracy.

P002 › NLP-Powered Contract Review System — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- BERT
- HuggingFace Transformers
- FastAPI
- React
- SharePoint

KEY METRICS & BUSINESS IMPACT

90% reduction in manual review time. Processes 500+ contracts/day with 94% clause classification accuracy.

3	2	Legal
Team Size	Duration	Domain

PROBLEM CONTEXT

Legal team spending 2,000+ hours/month manually reviewing vendor contracts for compliance clauses and risk flags.

P003

Real-Time Supply Chain Visibility Platform

Data Engineering

PROBLEM STATEMENT

No end-to-end visibility across 40+ suppliers, causing $8M in annual inventory write-offs and 15% on-time delivery failure.

SOLUTION APPROACH

Built event-driven data pipeline using Kafka for real-time supplier feeds. Created unified dashboard on Databricks with anomaly detection for delivery risk scoring.

PROJECT DETAILS

Domain	Manufacturing / Supply Chain
Team Size	6 people
Duration	5 months
Type	Data Engineering

TECHNOLOGY STACK

Apache Kafka · Databricks · Delta Lake · Python · Power BI · Azure

EXPECTED OUTCOMES

Reduced inventory write-offs by 60%, improved on-time delivery to 96%. ROI of 4x in Year 1.

P003 › Real-Time Supply Chain Visibility Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Apache Kafka
- Databricks
- Delta Lake
- Python
- Power BI
- Azure

KEY METRICS & BUSINESS IMPACT

Reduced inventory write-offs by 60%, improved on-time delivery to 96%. ROI of 4x in Year 1.

6	5	Manufacturing
Team Size	Duration	Domain

PROBLEM CONTEXT

No end-to-end visibility across 40+ suppliers, causing $8M in annual inventory write-offs and 15% on-time delivery failure.

P004

Healthcare Claims Fraud Detection

ML/AI

PROBLEM STATEMENT

Insurance provider losing $40M annually to fraudulent claims, with traditional rules-based system catching only 30% of fraud cases.

SOLUTION APPROACH

Graph neural network model analyzing claim relationships and provider networks. Combined with rule engine for explainability. Real-time scoring via Azure ML endpoints.

PROJECT DETAILS

Domain	Healthcare / Insurance
Team Size	5 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · PyTorch · Graph Neural Networks · Azure ML · Neo4j · SQL Server

EXPECTED OUTCOMES

Fraud detection rate improved from 30% to 78%. Estimated $18M annual savings with <2% false positive rate.

P004 › Healthcare Claims Fraud Detection — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch
- Graph Neural Networks
- Azure ML
- Neo4j
- SQL Server

KEY METRICS & BUSINESS IMPACT

Fraud detection rate improved from 30% to 78%. Estimated $18M annual savings with <2% false positive rate.

5	4	Healthcare
Team Size	Duration	Domain

PROBLEM CONTEXT

Insurance provider losing $40M annually to fraudulent claims, with traditional rules-based system catching only 30% of fraud cases.

P005

GenAI Document Intelligence for Pharma

NLP / GenAI

PROBLEM STATEMENT

Drug trial documentation spread across 10,000+ PDFs, making regulatory submission preparation extremely manual and error-prone.

SOLUTION APPROACH

RAG pipeline using GPT-4 and custom embeddings to answer regulatory queries. Built semantic search over clinical trial documents with source attribution.

PROJECT DETAILS

Domain	Pharma / Life Sciences
Team Size	4 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4 · LangChain · FAISS · Python · Azure Blob Storage · Streamlit

EXPECTED OUTCOMES

75% reduction in regulatory document preparation time. Passed FDA audit with zero documentation gaps.

P005 › GenAI Document Intelligence for Pharma — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4
- LangChain
- FAISS
- Python
- Azure Blob Storage
- Streamlit

KEY METRICS & BUSINESS IMPACT

75% reduction in regulatory document preparation time. Passed FDA audit with zero documentation gaps.

4	3	Pharma
Team Size	Duration	Domain

PROBLEM CONTEXT

Drug trial documentation spread across 10,000+ PDFs, making regulatory submission preparation extremely manual and error-prone.

P006

E-Commerce Recommendation Engine

ML/AI

PROBLEM STATEMENT

Low product discovery rate (2.1% click-through) on homepage and search, leading to missed upsell revenue.

SOLUTION APPROACH

Hybrid collaborative + content-based filtering model using user behavior, purchase history, and product embeddings. A/B tested via feature flags.

PROJECT DETAILS

Domain	Retail / E-Commerce
Team Size	5 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · TensorFlow · Redis · Apache Spark · AWS · Elasticsearch

EXPECTED OUTCOMES

CTR improved to 6.8% (3.2x baseline). $4.2M incremental revenue in Q1 post-launch.

P006 › E-Commerce Recommendation Engine — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- TensorFlow
- Redis
- Apache Spark
- AWS
- Elasticsearch

KEY METRICS & BUSINESS IMPACT

CTR improved to 6.8% (3.2x baseline). $4.2M incremental revenue in Q1 post-launch.

5	3	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

Low product discovery rate (2.1% click-through) on homepage and search, leading to missed upsell revenue.

P007

Predictive Maintenance for Industrial IoT

ML/AI

PROBLEM STATEMENT

Unplanned equipment downtime costing $500K/incident across 200 manufacturing assets with no predictive capability.

SOLUTION APPROACH

Time-series anomaly detection on sensor data streams using LSTM models. Edge deployment on Raspberry Pi devices with cloud aggregation on AWS IoT.

PROJECT DETAILS

Domain	Manufacturing / Energy
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · LSTM · TensorFlow Lite · AWS IoT · InfluxDB · Grafana

EXPECTED OUTCOMES

Reduced unplanned downtime by 67%. Saved $3.2M in Year 1 with 94% failure prediction accuracy (7-day horizon).

P007 › Predictive Maintenance for Industrial IoT — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LSTM
- TensorFlow Lite
- AWS IoT
- InfluxDB
- Grafana

KEY METRICS & BUSINESS IMPACT

Reduced unplanned downtime by 67%. Saved $3.2M in Year 1 with 94% failure prediction accuracy (7-day horizon).

4	4	Manufacturing
Team Size	Duration	Domain

PROBLEM CONTEXT

Unplanned equipment downtime costing $500K/incident across 200 manufacturing assets with no predictive capability.

P008

Automated Data Quality Framework

Data Engineering

PROBLEM STATEMENT

Data quality issues causing $1.5M in fines and 400 hours/month of manual data validation across 12 source systems.

SOLUTION APPROACH

Built Great Expectations-based framework with 500+ automated tests. Integrated into Airflow DAGs with Slack alerting and lineage tracking via dbt.

PROJECT DETAILS

Domain	Financial Services
Team Size	3 people
Duration	2 months
Type	Data Engineering

TECHNOLOGY STACK

Great Expectations · Apache Airflow · dbt · Python · Snowflake · Slack API

EXPECTED OUTCOMES

Zero data quality-related fines in 8 months post-launch. 95% reduction in manual validation effort.

P008 › Automated Data Quality Framework — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Great Expectations
- Apache Airflow
- dbt
- Python
- Snowflake
- Slack API

KEY METRICS & BUSINESS IMPACT

Zero data quality-related fines in 8 months post-launch. 95% reduction in manual validation effort.

3	2	Financial Services
Team Size	Duration	Domain

PROBLEM CONTEXT

Data quality issues causing $1.5M in fines and 400 hours/month of manual data validation across 12 source systems.

P009

Multi-Language Sentiment Analysis Platform

NLP / GenAI

PROBLEM STATEMENT

Brand monitoring team unable to process 500K+ daily social mentions in 12 languages, missing reputation crises in real time.

SOLUTION APPROACH

Multilingual BERT model fine-tuned on domain-specific data. Real-time pipeline via Kafka and Elasticsearch. Executive dashboard with crisis alerting.

PROJECT DETAILS

Domain	Consumer Goods / Marketing
Team Size	3 people
Duration	2 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · mBERT · HuggingFace · Apache Kafka · Elasticsearch · Kibana

EXPECTED OUTCOMES

Real-time monitoring across 12 languages with 89% sentiment accuracy. Detected 3 major brand crises within hours.

P009 › Multi-Language Sentiment Analysis Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- mBERT
- HuggingFace
- Apache Kafka
- Elasticsearch
- Kibana

KEY METRICS & BUSINESS IMPACT

Real-time monitoring across 12 languages with 89% sentiment accuracy. Detected 3 major brand crises within hours.

3	2	Consumer Goods
Team Size	Duration	Domain

PROBLEM CONTEXT

Brand monitoring team unable to process 500K+ daily social mentions in 12 languages, missing reputation crises in real time.

P010

Cloud Cost Optimization Intelligence

Data Engineering

PROBLEM STATEMENT

Cloud spend growing 40% YoY with no visibility into waste. $2.8M annual overspend identified by external audit.

SOLUTION APPROACH

Built cost attribution pipeline ingesting AWS Cost Explorer + Azure Cost Management. ML model predicts spend anomalies. FinOps dashboard with rightsizing recommendations.

PROJECT DETAILS

Domain	Technology / SaaS
Team Size	2 people
Duration	6 weeks
Type	Data Engineering

TECHNOLOGY STACK

Python · AWS Cost Explorer API · Azure Cost Management · Snowflake · Tableau · dbt

EXPECTED OUTCOMES

Identified $1.9M in annual savings. 34% reduction in cloud spend within 3 months of recommendations.

P010 › Cloud Cost Optimization Intelligence — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- AWS Cost Explorer API
- Azure Cost Management
- Snowflake
- Tableau
- dbt

KEY METRICS & BUSINESS IMPACT

Identified $1.9M in annual savings. 34% reduction in cloud spend within 3 months of recommendations.

2	6 weeks	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

Cloud spend growing 40% YoY with no visibility into waste. $2.8M annual overspend identified by external audit.

P011

LLM-Powered HR Policy Chatbot

NLP / GenAI

PROBLEM STATEMENT

HR team fielding 3,000+ repetitive policy questions/month, with 48-hour average response time frustrating employees.

SOLUTION APPROACH

RAG-based chatbot using GPT-3.5-turbo over internal HR policy documents. Integrated with Slack and MS Teams. Escalation routing to HR agents for complex queries.

PROJECT DETAILS

Domain	HR / Enterprise
Team Size	2 people
Duration	6 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-3.5 · LangChain · Pinecone · Python · Slack SDK · MS Teams API

EXPECTED OUTCOMES

Deflected 78% of HR queries. Average response time reduced from 48 hours to <30 seconds. 4.6/5 employee satisfaction.

P011 › LLM-Powered HR Policy Chatbot — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-3.5
- LangChain
- Pinecone
- Python
- Slack SDK
- MS Teams API

KEY METRICS & BUSINESS IMPACT

Deflected 78% of HR queries. Average response time reduced from 48 hours to <30 seconds. 4.6/5 employee satisfaction.

2	6 weeks	HR
Team Size	Duration	Domain

PROBLEM CONTEXT

HR team fielding 3,000+ repetitive policy questions/month, with 48-hour average response time frustrating employees.

P012

Real-Time Fraud Detection for Payments

ML/AI

PROBLEM STATEMENT

Payment processor experiencing 0.3% fraud rate ($9M annual losses) with 800ms legacy rule engine too slow for modern transaction volumes.

SOLUTION APPROACH

Streaming ML pipeline on Apache Flink with feature store on Redis. Gradient boosting model with 50ms inference latency. Shadow mode deployment for safe rollout.

PROJECT DETAILS

Domain	FinTech / Banking
Team Size	7 people
Duration	6 months
Type	ML/AI

TECHNOLOGY STACK

Apache Flink · Python · XGBoost · Redis · Apache Kafka · Kubernetes

EXPECTED OUTCOMES

Fraud rate reduced to 0.07%. $7.4M annual loss prevention. P99 latency of 45ms at 50K TPS.

P012 › Real-Time Fraud Detection for Payments — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Apache Flink
- Python
- XGBoost
- Redis
- Apache Kafka
- Kubernetes

KEY METRICS & BUSINESS IMPACT

Fraud rate reduced to 0.07%. $7.4M annual loss prevention. P99 latency of 45ms at 50K TPS.

7	6	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Payment processor experiencing 0.3% fraud rate ($9M annual losses) with 800ms legacy rule engine too slow for modern transaction volumes.

P013

Drug Discovery Target Identification

ML/AI

PROBLEM STATEMENT

Traditional target identification taking 3–5 years, blocking early-stage pipeline with $200M+ opportunity cost.

SOLUTION APPROACH

Graph ML model on protein interaction networks combined with literature mining via BioBERT. Prioritization scores for 10,000+ gene candidates.

PROJECT DETAILS

Domain	Pharma / Life Sciences
Team Size	6 people
Duration	8 months
Type	ML/AI

TECHNOLOGY STACK

Python · PyTorch Geometric · BioBERT · Neo4j · RDKit · AWS

EXPECTED OUTCOMES

Identified 3 novel targets validated in lab. Reduced initial screening time from 18 months to 6 weeks.

P013 › Drug Discovery Target Identification — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch Geometric
- BioBERT
- Neo4j
- RDKit
- AWS

KEY METRICS & BUSINESS IMPACT

Identified 3 novel targets validated in lab. Reduced initial screening time from 18 months to 6 weeks.

6	8	Pharma
Team Size	Duration	Domain

PROBLEM CONTEXT

Traditional target identification taking 3–5 years, blocking early-stage pipeline with $200M+ opportunity cost.

P014

Automated Code Review Assistant

NLP / GenAI

PROBLEM STATEMENT

Code review backlog causing 3-day average PR merge time, slowing release velocity and developer productivity.

SOLUTION APPROACH

LLM-based code review bot integrated with GitHub PRs. Fine-tuned CodeLlama model for security vulnerability detection and style compliance.

PROJECT DETAILS

Domain	Technology / DevOps
Team Size	3 people
Duration	2 months
Type	NLP / GenAI

TECHNOLOGY STACK

CodeLlama · Python · GitHub Actions · FastAPI · Docker · PostgreSQL

EXPECTED OUTCOMES

75% reduction in PR review time. Security vulnerability detection at 82% recall. 40% improvement in code quality scores.

P014 › Automated Code Review Assistant — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- CodeLlama
- Python
- GitHub Actions
- FastAPI
- Docker
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

75% reduction in PR review time. Security vulnerability detection at 82% recall. 40% improvement in code quality scores.

3	2	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

Code review backlog causing 3-day average PR merge time, slowing release velocity and developer productivity.

P015

Customer 360 Data Platform

Data Engineering

PROBLEM STATEMENT

Customer data siloed across 8 systems — CRM, billing, support, web analytics — preventing personalized experiences.

SOLUTION APPROACH

Built identity resolution pipeline using probabilistic matching. Unified customer profile in Snowflake with dbt transformation layer and real-time API for downstream apps.

PROJECT DETAILS

Domain	Retail / Banking
Team Size	5 people
Duration	4 months
Type	Data Engineering

TECHNOLOGY STACK

Snowflake · dbt · Apache Spark · Python · Airflow · REST API

EXPECTED OUTCOMES

Unified profiles for 12M customers. Enabled personalization that drove 18% uplift in cross-sell conversion.

P015 › Customer 360 Data Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Snowflake
- dbt
- Apache Spark
- Python
- Airflow
- REST API

KEY METRICS & BUSINESS IMPACT

Unified profiles for 12M customers. Enabled personalization that drove 18% uplift in cross-sell conversion.

5	4	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

Customer data siloed across 8 systems — CRM, billing, support, web analytics — preventing personalized experiences.

P016

NLP-Based Resume Screening System

NLP / GenAI

PROBLEM STATEMENT

Recruiting team manually screening 10,000+ resumes/month per role, with 72-hour time-to-shortlist frustrating hiring managers.

SOLUTION APPROACH

Sentence-BERT embeddings for semantic matching between JD and resume. Scoring pipeline with ATS integration. Bias audit module for fairness compliance.

PROJECT DETAILS

Domain	HR / Recruitment
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · Sentence-BERT · HuggingFace · FastAPI · Workday API · PostgreSQL

EXPECTED OUTCOMES

Time-to-shortlist reduced from 72 hours to 4 hours. 91% hiring manager satisfaction with shortlist quality.

P016 › NLP-Based Resume Screening System — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Sentence-BERT
- HuggingFace
- FastAPI
- Workday API
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Time-to-shortlist reduced from 72 hours to 4 hours. 91% hiring manager satisfaction with shortlist quality.

3	10 weeks	HR
Team Size	Duration	Domain

PROBLEM CONTEXT

Recruiting team manually screening 10,000+ resumes/month per role, with 72-hour time-to-shortlist frustrating hiring managers.

P017

Energy Demand Forecasting System

ML/AI

PROBLEM STATEMENT

Utility company over-procuring energy by 12% due to inaccurate demand forecasts, costing $15M in unnecessary contracts.

SOLUTION APPROACH

Ensemble forecasting: Prophet for trend + LSTM for short-term patterns + external regressors (weather, events). Deployed on Azure ML with automated daily retraining.

PROJECT DETAILS

Domain	Energy / Utilities
Team Size	4 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · Prophet · LSTM · Azure ML · SQL Server · Power BI

EXPECTED OUTCOMES

Forecast accuracy improved from 83% to 96%. Annual savings of $11M in energy procurement costs.

P017 › Energy Demand Forecasting System — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Prophet
- LSTM
- Azure ML
- SQL Server
- Power BI

KEY METRICS & BUSINESS IMPACT

Forecast accuracy improved from 83% to 96%. Annual savings of $11M in energy procurement costs.

4	3	Energy
Team Size	Duration	Domain

PROBLEM CONTEXT

Utility company over-procuring energy by 12% due to inaccurate demand forecasts, costing $15M in unnecessary contracts.

P018

Logistics Route Optimization Engine

ML/AI

PROBLEM STATEMENT

Fleet of 500 vehicles using static route plans, resulting in 22% excess mileage and $6M annual fuel waste.

SOLUTION APPROACH

Dynamic vehicle routing with Google OR-Tools. Real-time traffic integration via HERE API. Constraint satisfaction for time windows, capacity, and driver hours.

PROJECT DETAILS

Domain	Logistics / Supply Chain
Team Size	5 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · Google OR-Tools · HERE API · FastAPI · PostgreSQL · React

EXPECTED OUTCOMES

18% reduction in total mileage. $4.8M annual fuel savings. On-time delivery improved from 84% to 97%.

P018 › Logistics Route Optimization Engine — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Google OR-Tools
- HERE API
- FastAPI
- PostgreSQL
- React

KEY METRICS & BUSINESS IMPACT

18% reduction in total mileage. $4.8M annual fuel savings. On-time delivery improved from 84% to 97%.

5	4	Logistics
Team Size	Duration	Domain

PROBLEM CONTEXT

Fleet of 500 vehicles using static route plans, resulting in 22% excess mileage and $6M annual fuel waste.

P019

Social Media Analytics Pipeline

Data Engineering

PROBLEM STATEMENT

Marketing team running weekly batch analytics on social data, missing real-time campaign performance signals.

SOLUTION APPROACH

Streaming ingestion from Twitter/Instagram APIs via Kafka. Spark Streaming for aggregations. Near-real-time Power BI dashboard refreshing every 5 minutes.

PROJECT DETAILS

Domain	Marketing / Media
Team Size	3 people
Duration	2 months
Type	Data Engineering

TECHNOLOGY STACK

Apache Kafka · Spark Streaming · Python · Twitter API · Azure · Power BI

EXPECTED OUTCOMES

Campaign optimization latency reduced from 7 days to 15 minutes. 28% improvement in campaign ROI.

P019 › Social Media Analytics Pipeline — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Apache Kafka
- Spark Streaming
- Python
- Twitter API
- Azure
- Power BI

KEY METRICS & BUSINESS IMPACT

Campaign optimization latency reduced from 7 days to 15 minutes. 28% improvement in campaign ROI.

3	2	Marketing
Team Size	Duration	Domain

PROBLEM CONTEXT

Marketing team running weekly batch analytics on social data, missing real-time campaign performance signals.

P020

Medical Image Classification for Radiology

ML/AI

PROBLEM STATEMENT

Radiologists reviewing 400+ chest X-rays daily with 48-hour report turnaround and 12% missed finding rate.

SOLUTION APPROACH

EfficientNet-based CNN fine-tuned on 120K labeled X-rays for 14 pathology classes. Integrated with PACS via DICOM standard. Uncertainty quantification for triage.

PROJECT DETAILS

Domain	Healthcare
Team Size	6 people
Duration	6 months
Type	ML/AI

TECHNOLOGY STACK

Python · PyTorch · EfficientNet · DICOM · FastAPI · AWS

EXPECTED OUTCOMES

Turnaround reduced to 8 hours. Sensitivity improved to 94% (from 88%). FDA 510(k) clearance obtained.

P020 › Medical Image Classification for Radiology — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch
- EfficientNet
- DICOM
- FastAPI
- AWS

KEY METRICS & BUSINESS IMPACT

Turnaround reduced to 8 hours. Sensitivity improved to 94% (from 88%). FDA 510(k) clearance obtained.

6	6	Healthcare
Team Size	Duration	Domain

PROBLEM CONTEXT

Radiologists reviewing 400+ chest X-rays daily with 48-hour report turnaround and 12% missed finding rate.

P021

Automated Financial Reporting Pipeline

Data Engineering

PROBLEM STATEMENT

Month-end close taking 12 days with 60% time spent on manual data consolidation across 30 Excel files.

SOLUTION APPROACH

Automated ETL from ERP systems into Snowflake. dbt models for financial consolidation logic. Pre-built report templates in Power BI with row-level security.

PROJECT DETAILS

Domain	Financial Services
Team Size	3 people
Duration	3 months
Type	Data Engineering

TECHNOLOGY STACK

Snowflake · dbt · Airflow · SAP ERP API · Power BI · Python

EXPECTED OUTCOMES

Month-end close reduced from 12 days to 3 days. Eliminated 95% of manual effort. Zero restatements in 12 months.

P021 › Automated Financial Reporting Pipeline — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Snowflake
- dbt
- Airflow
- SAP ERP API
- Power BI
- Python

KEY METRICS & BUSINESS IMPACT

Month-end close reduced from 12 days to 3 days. Eliminated 95% of manual effort. Zero restatements in 12 months.

3	3	Financial Services
Team Size	Duration	Domain

PROBLEM CONTEXT

Month-end close taking 12 days with 60% time spent on manual data consolidation across 30 Excel files.

P022

Conversational AI for Banking Support

NLP / GenAI

PROBLEM STATEMENT

Call center handling 50,000 monthly inquiries at $8 average cost per call. 70% of calls are repetitive balance/transaction queries.

SOLUTION APPROACH

Rasa-based conversational AI with banking core system integration. Multi-turn dialogue for account queries, fund transfers. Live agent handoff with conversation context.

PROJECT DETAILS

Domain	Banking / FinTech
Team Size	5 people
Duration	5 months
Type	NLP / GenAI

TECHNOLOGY STACK

Rasa · Python · Core Banking API · Redis · Docker · Kubernetes

EXPECTED OUTCOMES

Automated 65% of call volume. Annual savings of $2.8M. CSAT maintained at 4.2/5.

P022 › Conversational AI for Banking Support — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Rasa
- Python
- Core Banking API
- Redis
- Docker
- Kubernetes

KEY METRICS & BUSINESS IMPACT

Automated 65% of call volume. Annual savings of $2.8M. CSAT maintained at 4.2/5.

5	5	Banking
Team Size	Duration	Domain

PROBLEM CONTEXT

Call center handling 50,000 monthly inquiries at $8 average cost per call. 70% of calls are repetitive balance/transaction queries.

P023

Product Catalog Embedding & Search

NLP / GenAI

PROBLEM STATEMENT

Keyword-based search returning irrelevant results for 35% of queries, leading to $1.2M monthly in abandoned searches.

SOLUTION APPROACH

Product embeddings using fine-tuned CLIP model for text+image. Semantic search index in Elasticsearch with KNN plugin. A/B testing framework for ranking experiments.

PROJECT DETAILS

Domain	E-Commerce / Retail
Team Size	4 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · CLIP · Elasticsearch · FastAPI · AWS S3 · React

EXPECTED OUTCOMES

Search relevance improved by 47%. Abandoned search rate reduced to 11%. $680K monthly revenue recovery.

P023 › Product Catalog Embedding & Search — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- CLIP
- Elasticsearch
- FastAPI
- AWS S3
- React

KEY METRICS & BUSINESS IMPACT

Search relevance improved by 47%. Abandoned search rate reduced to 11%. $680K monthly revenue recovery.

4	3	E-Commerce
Team Size	Duration	Domain

PROBLEM CONTEXT

Keyword-based search returning irrelevant results for 35% of queries, leading to $1.2M monthly in abandoned searches.

P024

Real-Time Patient Deterioration Alerts

ML/AI

PROBLEM STATEMENT

ICU team missing early deterioration signals, with 22% of preventable codes occurring 6+ hours after first warning signs in vitals.

SOLUTION APPROACH

LSTM model on continuous vitals stream predicts deterioration 6 hours ahead. Alert system integrated with nurse station monitors via HL7 FHIR.

PROJECT DETAILS

Domain	Healthcare
Team Size	5 people
Duration	5 months
Type	ML/AI

TECHNOLOGY STACK

Python · LSTM · TensorFlow · HL7 FHIR · Apache Kafka · PostgreSQL

EXPECTED OUTCOMES

Early warning rate improved by 41%. Preventable deterioration events reduced by 35%. IRB-approved clinical trial underway.

P024 › Real-Time Patient Deterioration Alerts — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LSTM
- TensorFlow
- HL7 FHIR
- Apache Kafka
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Early warning rate improved by 41%. Preventable deterioration events reduced by 35%. IRB-approved clinical trial underway.

5	5	Healthcare
Team Size	Duration	Domain

PROBLEM CONTEXT

ICU team missing early deterioration signals, with 22% of preventable codes occurring 6+ hours after first warning signs in vitals.

P025

Data Lakehouse Migration (On-Prem to Cloud)

Data Engineering

PROBLEM STATEMENT

Legacy on-prem Hadoop cluster (500TB) reaching end-of-life, costing $1.2M/year to maintain with 8-hour SLA for critical reports.

SOLUTION APPROACH

Lift-and-shift migration to Azure Data Lake Gen2 + Databricks. Delta Lake for ACID transactions. Spark job optimization reduced compute costs by 45%.

PROJECT DETAILS

Domain	Retail
Team Size	8 people
Duration	7 months
Type	Data Engineering

TECHNOLOGY STACK

Azure Data Lake Gen2 · Databricks · Delta Lake · Apache Spark · dbt · Terraform

EXPECTED OUTCOMES

Infrastructure costs reduced by 55%. Report SLA improved from 8 hours to 90 minutes. Zero data loss during migration.

P025 › Data Lakehouse Migration (On-Prem to Cloud) — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Azure Data Lake Gen2
- Databricks
- Delta Lake
- Apache Spark
- dbt
- Terraform

KEY METRICS & BUSINESS IMPACT

Infrastructure costs reduced by 55%. Report SLA improved from 8 hours to 90 minutes. Zero data loss during migration.

8	7	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

Legacy on-prem Hadoop cluster (500TB) reaching end-of-life, costing $1.2M/year to maintain with 8-hour SLA for critical reports.

P026

LLM-Assisted Sales Proposal Generator

NLP / GenAI

PROBLEM STATEMENT

Sales team spending 6–8 hours per proposal, with inconsistent messaging causing 40% loss rate on qualified deals.

SOLUTION APPROACH

RAG system over past winning proposals + product catalog. GPT-4 generates first-draft proposals customized to prospect industry and deal size.

PROJECT DETAILS

Domain	B2B SaaS / Sales
Team Size	2 people
Duration	6 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4 · LangChain · Pinecone · Python · Salesforce API · React

EXPECTED OUTCOMES

Proposal creation time reduced from 8 hours to 45 minutes. Win rate improved by 18% within 2 quarters.

P026 › LLM-Assisted Sales Proposal Generator — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4
- LangChain
- Pinecone
- Python
- Salesforce API
- React

KEY METRICS & BUSINESS IMPACT

Proposal creation time reduced from 8 hours to 45 minutes. Win rate improved by 18% within 2 quarters.

2	6 weeks	B2B SaaS
Team Size	Duration	Domain

PROBLEM CONTEXT

Sales team spending 6–8 hours per proposal, with inconsistent messaging causing 40% loss rate on qualified deals.

P027

Inventory Demand Forecasting at SKU Level

ML/AI

PROBLEM STATEMENT

28% stockout rate and 19% overstock causing $22M in lost sales and $8M in write-offs annually across 50,000 SKUs.

SOLUTION APPROACH

Hierarchical time-series forecasting using M5 winning approach (LightGBM + reconciliation). Integrated with SAP inventory for automated replenishment triggers.

PROJECT DETAILS

Domain	Retail / FMCG
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · LightGBM · Statsmodels · SAP API · Azure · Snowflake

EXPECTED OUTCOMES

Stockout rate reduced to 9%. Overstock write-offs down 62%. $18M combined improvement in first year.

P027 › Inventory Demand Forecasting at SKU Level — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LightGBM
- Statsmodels
- SAP API
- Azure
- Snowflake

KEY METRICS & BUSINESS IMPACT

Stockout rate reduced to 9%. Overstock write-offs down 62%. $18M combined improvement in first year.

4	4	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

28% stockout rate and 19% overstock causing $22M in lost sales and $8M in write-offs annually across 50,000 SKUs.

P028

Automated Compliance Document Extraction

NLP / GenAI

PROBLEM STATEMENT

Compliance team manually extracting data from 2,000+ regulatory filings/month, with 5% error rate causing audit findings.

SOLUTION APPROACH

Document AI pipeline using Google Document AI + custom NER model for entity extraction. Human-in-the-loop review for low-confidence extractions.

PROJECT DETAILS

Domain	Financial Services / Compliance
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · Google Document AI · spaCy · FastAPI · Google Cloud · PostgreSQL

EXPECTED OUTCOMES

Manual effort reduced by 85%. Error rate reduced to 0.3%. Compliance audit findings eliminated for 2 consecutive years.

P028 › Automated Compliance Document Extraction — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Google Document AI
- spaCy
- FastAPI
- Google Cloud
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Manual effort reduced by 85%. Error rate reduced to 0.3%. Compliance audit findings eliminated for 2 consecutive years.

3	10 weeks	Financial Services
Team Size	Duration	Domain

PROBLEM CONTEXT

Compliance team manually extracting data from 2,000+ regulatory filings/month, with 5% error rate causing audit findings.

P029

Cybersecurity Threat Intelligence Platform

ML/AI

PROBLEM STATEMENT

SOC team ingesting 5M+ daily events with 98% false positive rate, causing alert fatigue and 4-hour mean detection time.

SOLUTION APPROACH

ML-based alert triage using isolation forest + UEBA behavioral models. Knowledge graph for threat correlation. SOAR integration for automated response playbooks.

PROJECT DETAILS

Domain	Cybersecurity
Team Size	5 people
Duration	5 months
Type	ML/AI

TECHNOLOGY STACK

Python · Splunk · Neo4j · Scikit-learn · Elasticsearch · Docker

EXPECTED OUTCOMES

False positive rate reduced to 12%. Mean detection time improved from 4 hours to 18 minutes. SOC capacity freed by 60%.

P029 › Cybersecurity Threat Intelligence Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Splunk
- Neo4j
- Scikit-learn
- Elasticsearch
- Docker

KEY METRICS & BUSINESS IMPACT

False positive rate reduced to 12%. Mean detection time improved from 4 hours to 18 minutes. SOC capacity freed by 60%.

5	5	Cybersecurity
Team Size	Duration	Domain

PROBLEM CONTEXT

SOC team ingesting 5M+ daily events with 98% false positive rate, causing alert fatigue and 4-hour mean detection time.

P030

Warehouse Slotting Optimization

ML/AI

PROBLEM STATEMENT

Warehouse picking efficiency at 65 lines/hour vs. industry benchmark of 120, due to poor product placement strategy.

SOLUTION APPROACH

ML-based slotting engine using order co-occurrence mining and velocity analysis. Genetic algorithm for optimal bin assignment with travel-time minimization.

PROJECT DETAILS

Domain	Logistics / Warehouse
Team Size	3 people
Duration	2 months
Type	ML/AI

TECHNOLOGY STACK

Python · Scikit-learn · DEAP (genetic algo) · PostgreSQL · FastAPI · Power BI

EXPECTED OUTCOMES

Picking rate improved to 108 lines/hour (66% improvement). Annual labor savings of $1.4M.

P030 › Warehouse Slotting Optimization — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Scikit-learn
- DEAP (genetic algo)
- PostgreSQL
- FastAPI
- Power BI

KEY METRICS & BUSINESS IMPACT

Picking rate improved to 108 lines/hour (66% improvement). Annual labor savings of $1.4M.

3	2	Logistics
Team Size	Duration	Domain

PROBLEM CONTEXT

Warehouse picking efficiency at 65 lines/hour vs. industry benchmark of 120, due to poor product placement strategy.

P031

Streaming CDC Data Pipeline

Data Engineering

PROBLEM STATEMENT

Batch ETL causing 4-hour data lag in analytics platform, making real-time risk dashboards impossible.

SOLUTION APPROACH

Debezium CDC from PostgreSQL source to Kafka topics. Kafka Streams for transformation. Apache Iceberg on S3 for ACID lakehouse with sub-minute latency.

PROJECT DETAILS

Domain	FinTech / Data Platform
Team Size	4 people
Duration	3 months
Type	Data Engineering

TECHNOLOGY STACK

Debezium · Apache Kafka · Kafka Streams · Apache Iceberg · AWS S3 · Trino

EXPECTED OUTCOMES

Data latency reduced from 4 hours to <2 minutes. Enabled real-time risk monitoring across 8 downstream dashboards.

P031 › Streaming CDC Data Pipeline — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Debezium
- Apache Kafka
- Kafka Streams
- Apache Iceberg
- AWS S3
- Trino

KEY METRICS & BUSINESS IMPACT

Data latency reduced from 4 hours to <2 minutes. Enabled real-time risk monitoring across 8 downstream dashboards.

4	3	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Batch ETL causing 4-hour data lag in analytics platform, making real-time risk dashboards impossible.

P032

AI-Powered Quality Control for Manufacturing

ML/AI

PROBLEM STATEMENT

Manual visual inspection catching only 72% of defects, with $3.5M annual cost from defective products reaching customers.

SOLUTION APPROACH

Computer vision system using YOLOv8 trained on 50K labeled images. Edge deployment on NVIDIA Jetson for real-time line inspection at 200ms latency.

PROJECT DETAILS

Domain	Manufacturing
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · YOLOv8 · NVIDIA Jetson · OpenCV · FastAPI · InfluxDB

EXPECTED OUTCOMES

Defect detection rate improved from 72% to 97.3%. Customer complaints reduced by 89%. $2.9M annual savings.

P032 › AI-Powered Quality Control for Manufacturing — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- YOLOv8
- NVIDIA Jetson
- OpenCV
- FastAPI
- InfluxDB

KEY METRICS & BUSINESS IMPACT

Defect detection rate improved from 72% to 97.3%. Customer complaints reduced by 89%. $2.9M annual savings.

4	4	Manufacturing
Team Size	Duration	Domain

PROBLEM CONTEXT

Manual visual inspection catching only 72% of defects, with $3.5M annual cost from defective products reaching customers.

P033

Student Performance Prediction System

ML/AI

PROBLEM STATEMENT

University losing 15% of students annually to dropout, with interventions happening too late (after grades posted).

SOLUTION APPROACH

Early warning system using LMS engagement data, attendance, and assessment scores. Random Forest model predicts at-risk students 6 weeks before exams.

PROJECT DETAILS

Domain	EdTech
Team Size	2 people
Duration	6 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · Scikit-learn · Random Forest · Canvas LMS API · PostgreSQL · React

EXPECTED OUTCOMES

At-risk students identified 6 weeks earlier. Dropout rate reduced by 28% after advisor intervention program launch.

P033 › Student Performance Prediction System — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Scikit-learn
- Random Forest
- Canvas LMS API
- PostgreSQL
- React

KEY METRICS & BUSINESS IMPACT

At-risk students identified 6 weeks earlier. Dropout rate reduced by 28% after advisor intervention program launch.

2	6 weeks	EdTech
Team Size	Duration	Domain

PROBLEM CONTEXT

University losing 15% of students annually to dropout, with interventions happening too late (after grades posted).

P034

Knowledge Graph for Enterprise Search

NLP / GenAI

PROBLEM STATEMENT

Enterprise search returning irrelevant results with no understanding of entity relationships across 500TB of internal documents.

SOLUTION APPROACH

Entity extraction pipeline using spaCy + custom models. Knowledge graph in Neo4j linking people, projects, clients, and technologies. GraphQL API for search.

PROJECT DETAILS

Domain	Technology / Enterprise
Team Size	5 people
Duration	4 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · spaCy · Neo4j · GraphQL · Elasticsearch · Azure

EXPECTED OUTCOMES

Search relevance score improved by 58%. Employee time finding information reduced from 2.5 hours/day to 45 minutes.

P034 › Knowledge Graph for Enterprise Search — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- spaCy
- Neo4j
- GraphQL
- Elasticsearch
- Azure

KEY METRICS & BUSINESS IMPACT

Search relevance score improved by 58%. Employee time finding information reduced from 2.5 hours/day to 45 minutes.

5	4	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

Enterprise search returning irrelevant results with no understanding of entity relationships across 500TB of internal documents.

P035

Loan Default Risk Scoring Engine

ML/AI

PROBLEM STATEMENT

Credit risk model built in 2018 showing 15% performance degradation, with NPL ratio rising to 4.2% vs. industry 2.1%.

SOLUTION APPROACH

Rebuilt with XGBoost on enriched features including alternative data (utility payments, social). Fairness constraints via Fairlearn. Shadow deployment with champion/challenger.

PROJECT DETAILS

Domain	Banking / FinTech
Team Size	4 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · XGBoost · Fairlearn · MLflow · Azure ML · SQL Server

EXPECTED OUTCOMES

NPL ratio reduced to 1.8%. Model AUC improved from 0.71 to 0.88. Passed RBI model governance audit.

P035 › Loan Default Risk Scoring Engine — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- XGBoost
- Fairlearn
- MLflow
- Azure ML
- SQL Server

KEY METRICS & BUSINESS IMPACT

NPL ratio reduced to 1.8%. Model AUC improved from 0.71 to 0.88. Passed RBI model governance audit.

4	3	Banking
Team Size	Duration	Domain

PROBLEM CONTEXT

Credit risk model built in 2018 showing 15% performance degradation, with NPL ratio rising to 4.2% vs. industry 2.1%.

P036

Real-Time Pricing Engine for E-Commerce

ML/AI

PROBLEM STATEMENT

Static pricing updated weekly, leaving $8M in annual margin on the table vs. competitors using dynamic pricing.

SOLUTION APPROACH

Reinforcement learning pricing agent trained in simulation. Deployed with guardrails for margin floors. A/B tested via Optimizely with 15-minute price update cadence.

PROJECT DETAILS

Domain	E-Commerce / Retail
Team Size	4 people
Duration	5 months
Type	ML/AI

TECHNOLOGY STACK

Python · Stable Baselines3 · Redis · FastAPI · Optimizely · Snowflake

EXPECTED OUTCOMES

Gross margin improved by 3.2 percentage points. $6.4M annual incremental margin. 99.9% uptime on 10M daily price decisions.

P036 › Real-Time Pricing Engine for E-Commerce — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Stable Baselines3
- Redis
- FastAPI
- Optimizely
- Snowflake

KEY METRICS & BUSINESS IMPACT

Gross margin improved by 3.2 percentage points. $6.4M annual incremental margin. 99.9% uptime on 10M daily price decisions.

4	5	E-Commerce
Team Size	Duration	Domain

PROBLEM CONTEXT

Static pricing updated weekly, leaving $8M in annual margin on the table vs. competitors using dynamic pricing.

P037

Multi-Modal Product Content Generation

NLP / GenAI

PROBLEM STATEMENT

Content team creating product descriptions for 100K+ catalog items, taking 8 minutes per product at $3 per description.

SOLUTION APPROACH

Fine-tuned GPT-4 pipeline generating descriptions from product images + specifications. Human review workflow for brand voice compliance. Integrated with PIM system.

PROJECT DETAILS

Domain	E-Commerce / Marketing
Team Size	2 people
Duration	5 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4V · Python · FastAPI · Akeneo PIM API · AWS Lambda · PostgreSQL

EXPECTED OUTCOMES

Cost per description reduced from $3 to $0.08. Processing time from 8 minutes to 12 seconds. 96% brand compliance rate.

P037 › Multi-Modal Product Content Generation — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4V
- Python
- FastAPI
- Akeneo PIM API
- AWS Lambda
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Cost per description reduced from $3 to $0.08. Processing time from 8 minutes to 12 seconds. 96% brand compliance rate.

2	5 weeks	E-Commerce
Team Size	Duration	Domain

PROBLEM CONTEXT

Content team creating product descriptions for 100K+ catalog items, taking 8 minutes per product at $3 per description.

P038

Data Mesh Implementation for Telco

Data Engineering

PROBLEM STATEMENT

Central data team bottleneck causing 3-week average SLA for new data products, blocking 8 business domains.

SOLUTION APPROACH

Data Mesh architecture with domain-owned data products on GCP. Dataplex for governance. Data product contracts via OpenAPI. Central self-serve infrastructure platform.

PROJECT DETAILS

Domain	Telecom / Data Platform
Team Size	10 people
Duration	9 months
Type	Data Engineering

TECHNOLOGY STACK

GCP Dataplex · BigQuery · dbt · Apache Kafka · Terraform · Python

EXPECTED OUTCOMES

New data product SLA reduced from 3 weeks to 3 days. 40+ data products launched by domain teams in first 6 months.

P038 › Data Mesh Implementation for Telco — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- GCP Dataplex
- BigQuery
- dbt
- Apache Kafka
- Terraform
- Python

KEY METRICS & BUSINESS IMPACT

New data product SLA reduced from 3 weeks to 3 days. 40+ data products launched by domain teams in first 6 months.

10	9	Telecom
Team Size	Duration	Domain

PROBLEM CONTEXT

Central data team bottleneck causing 3-week average SLA for new data products, blocking 8 business domains.

P039

Satellite Imagery Analysis for Agriculture

ML/AI

PROBLEM STATEMENT

Agricultural insurer manually assessing crop damage claims, with 45-day settlement time and 8% fraudulent claim rate.

SOLUTION APPROACH

CNN model analyzing Sentinel-2 satellite imagery for crop stress detection. Change detection pre/post-event for loss quantification. Automated claim pre-assessment.

PROJECT DETAILS

Domain	Agriculture / GeoSpatial
Team Size	5 people
Duration	6 months
Type	ML/AI

TECHNOLOGY STACK

Python · PyTorch · Sentinel-2 API · GDAL · Google Earth Engine · PostgreSQL

EXPECTED OUTCOMES

Claim settlement time reduced from 45 days to 5 days. Fraud detection at 76% accuracy. $12M in fraudulent claims prevented.

P039 › Satellite Imagery Analysis for Agriculture — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch
- Sentinel-2 API
- GDAL
- Google Earth Engine
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Claim settlement time reduced from 45 days to 5 days. Fraud detection at 76% accuracy. $12M in fraudulent claims prevented.

5	6	Agriculture
Team Size	Duration	Domain

PROBLEM CONTEXT

Agricultural insurer manually assessing crop damage claims, with 45-day settlement time and 8% fraudulent claim rate.

P040

Voice-of-Customer Analytics Platform

NLP / GenAI

PROBLEM STATEMENT

Customer feedback arriving through 6 channels (email, chat, NPS, reviews, calls, social) with no unified analysis capability.

SOLUTION APPROACH

Unified NLP pipeline with topic modeling (BERTopic), sentiment analysis, and complaint classification. Executive CX dashboard with drill-down to verbatim.

PROJECT DETAILS

Domain	Consumer Goods / CX
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · BERTopic · HuggingFace · Apache Kafka · Elasticsearch · Tableau

EXPECTED OUTCOMES

360° customer view across 6 channels. Identified top 3 product issues driving 40% of negative NPS. NPS improved 12 points in 6 months.

P040 › Voice-of-Customer Analytics Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- BERTopic
- HuggingFace
- Apache Kafka
- Elasticsearch
- Tableau

KEY METRICS & BUSINESS IMPACT

360° customer view across 6 channels. Identified top 3 product issues driving 40% of negative NPS. NPS improved 12 points in 6 months.

3	10 weeks	Consumer Goods
Team Size	Duration	Domain

PROBLEM CONTEXT

Customer feedback arriving through 6 channels (email, chat, NPS, reviews, calls, social) with no unified analysis capability.

P041

Blockchain-Backed Supply Chain Traceability

Data Engineering

PROBLEM STATEMENT

Brand facing regulatory scrutiny on food sourcing claims with no immutable audit trail across 200+ global suppliers.

SOLUTION APPROACH

Hyperledger Fabric private blockchain for provenance tracking. IoT sensors at key checkpoints. QR code consumer-facing transparency portal.

PROJECT DETAILS

Domain	Food & Beverage / Supply Chain
Team Size	6 people
Duration	7 months
Type	Data Engineering

TECHNOLOGY STACK

Hyperledger Fabric · Node.js · Python · IoT Sensors · React · AWS

EXPECTED OUTCOMES

Full farm-to-shelf traceability for 85% of SKUs. Passed two FDA and EU regulatory audits. Consumer trust score +22 points.

P041 › Blockchain-Backed Supply Chain Traceability — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Hyperledger Fabric
- Node.js
- Python
- IoT Sensors
- React
- AWS

KEY METRICS & BUSINESS IMPACT

Full farm-to-shelf traceability for 85% of SKUs. Passed two FDA and EU regulatory audits. Consumer trust score +22 points.

6	7	Food & Beverage
Team Size	Duration	Domain

PROBLEM CONTEXT

Brand facing regulatory scrutiny on food sourcing claims with no immutable audit trail across 200+ global suppliers.

P042

Automated Algorithmic Trading Backtester

Data Engineering

PROBLEM STATEMENT

Quant team spending 3 weeks per strategy backtest due to fragmented tooling and no vectorized computation layer.

SOLUTION APPROACH

High-performance backtesting engine using Numba-accelerated vectorized operations. Walk-forward validation, transaction cost modeling, risk metrics library.

PROJECT DETAILS

Domain	FinTech / Capital Markets
Team Size	3 people
Duration	2 months
Type	Data Engineering

TECHNOLOGY STACK

Python · Numba · Pandas · Redis · FastAPI · InfluxDB

EXPECTED OUTCOMES

Backtest time reduced from 3 weeks to 4 hours. Enabled 10x more strategy iterations per quarter. 3 live strategies launched.

P042 › Automated Algorithmic Trading Backtester — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Numba
- Pandas
- Redis
- FastAPI
- InfluxDB

KEY METRICS & BUSINESS IMPACT

Backtest time reduced from 3 weeks to 4 hours. Enabled 10x more strategy iterations per quarter. 3 live strategies launched.

3	2	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Quant team spending 3 weeks per strategy backtest due to fragmented tooling and no vectorized computation layer.

P043

AI Tutor for Adaptive Learning

NLP / GenAI

PROBLEM STATEMENT

Online learning platform with 20% course completion rate due to one-size-fits-all content delivery and no personalized feedback.

SOLUTION APPROACH

Knowledge tracing using Deep Knowledge Tracing (DKT) to model student mastery. LLM for personalized hint generation. Adaptive content sequencing engine.

PROJECT DETAILS

Domain	EdTech
Team Size	4 people
Duration	4 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · PyTorch · OpenAI API · FastAPI · React · PostgreSQL

EXPECTED OUTCOMES

Course completion rate improved from 20% to 51%. Assessment scores improved 23% on average. NPS score: 72.

P043 › AI Tutor for Adaptive Learning — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch
- OpenAI API
- FastAPI
- React
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Course completion rate improved from 20% to 51%. Assessment scores improved 23% on average. NPS score: 72.

4	4	EdTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Online learning platform with 20% course completion rate due to one-size-fits-all content delivery and no personalized feedback.

P044

Real-Time Sports Analytics Platform

ML/AI

PROBLEM STATEMENT

Broadcasting team using 48-hour delayed match statistics, unable to deliver live data-driven insights during telecasts.

SOLUTION APPROACH

Computer vision tracking system using YOLOv8 for player/ball tracking. Real-time event detection pipeline. Low-latency broadcast graphics integration.

PROJECT DETAILS

Domain	Sports / Media
Team Size	6 people
Duration	6 months
Type	ML/AI

TECHNOLOGY STACK

Python · YOLOv8 · OpenCV · Apache Kafka · FastAPI · WebSocket

EXPECTED OUTCOMES

Live statistics with <500ms latency. Player tracking accuracy 94%. Integrated with 3 major broadcast networks.

P044 › Real-Time Sports Analytics Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- YOLOv8
- OpenCV
- Apache Kafka
- FastAPI
- WebSocket

KEY METRICS & BUSINESS IMPACT

Live statistics with <500ms latency. Player tracking accuracy 94%. Integrated with 3 major broadcast networks.

6	6	Sports
Team Size	Duration	Domain

PROBLEM CONTEXT

Broadcasting team using 48-hour delayed match statistics, unable to deliver live data-driven insights during telecasts.

P045

Enterprise Knowledge Base with RAG

NLP / GenAI

PROBLEM STATEMENT

Technical support team answering same 500 questions repeatedly, with new engineers taking 3 months to become productive.

SOLUTION APPROACH

RAG system indexing 10K+ internal documents (wikis, SOPs, tickets). Semantic chunking with parent-child retrieval. Confidence scoring and human escalation.

PROJECT DETAILS

Domain	Technology / Enterprise
Team Size	3 people
Duration	2 months
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4 · LlamaIndex · Weaviate · Python · Confluence API · React

EXPECTED OUTCOMES

Support ticket deflection at 71%. New engineer ramp-up time reduced from 3 months to 5 weeks. $890K annual support cost savings.

P045 › Enterprise Knowledge Base with RAG — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4
- LlamaIndex
- Weaviate
- Python
- Confluence API
- React

KEY METRICS & BUSINESS IMPACT

Support ticket deflection at 71%. New engineer ramp-up time reduced from 3 months to 5 weeks. $890K annual support cost savings.

3	2	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

Technical support team answering same 500 questions repeatedly, with new engineers taking 3 months to become productive.

P046

Patient Readmission Risk Prediction

ML/AI

PROBLEM STATEMENT

Hospital facing CMS penalties for 30-day readmission rates 1.8% above national benchmark, risking $4.2M in reimbursement reduction.

SOLUTION APPROACH

Ensemble model (LightGBM + logistic regression for interpretability) using EHR data. SHAP explanations for clinician trust. Daily risk score integration with discharge workflow.

PROJECT DETAILS

Domain	Healthcare
Team Size	4 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · LightGBM · SHAP · FHIR API · PostgreSQL · Tableau

EXPECTED OUTCOMES

Readmission rate reduced 22%. Avoided $3.1M in CMS penalties. Adopted across 4 hospital campuses.

P046 › Patient Readmission Risk Prediction — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LightGBM
- SHAP
- FHIR API
- PostgreSQL
- Tableau

KEY METRICS & BUSINESS IMPACT

Readmission rate reduced 22%. Avoided $3.1M in CMS penalties. Adopted across 4 hospital campuses.

4	3	Healthcare
Team Size	Duration	Domain

PROBLEM CONTEXT

Hospital facing CMS penalties for 30-day readmission rates 1.8% above national benchmark, risking $4.2M in reimbursement reduction.

P047

Geospatial Crime Prediction Platform

ML/AI

PROBLEM STATEMENT

Police department reactive to crime with no data-driven patrol allocation, resulting in 20% under-patrolling of high-risk areas.

SOLUTION APPROACH

Spatio-temporal ML model using historical crime data, socioeconomic features, and weather. Risk heatmap updated every 4 hours. Privacy-preserving design reviewed by ethics board.

PROJECT DETAILS

Domain	Public Safety / Government
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · GeoPandas · XGBoost · PostGIS · Folium · FastAPI

EXPECTED OUTCOMES

High-risk area patrol coverage improved by 35%. Property crime reduced 14% in pilot zones. Ethics board approved.

P047 › Geospatial Crime Prediction Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- GeoPandas
- XGBoost
- PostGIS
- Folium
- FastAPI

KEY METRICS & BUSINESS IMPACT

High-risk area patrol coverage improved by 35%. Property crime reduced 14% in pilot zones. Ethics board approved.

4	4	Public Safety
Team Size	Duration	Domain

PROBLEM CONTEXT

Police department reactive to crime with no data-driven patrol allocation, resulting in 20% under-patrolling of high-risk areas.

P048

Automated Invoice Processing

NLP / GenAI

PROBLEM STATEMENT

AP team manually processing 15,000 invoices/month at $8 per invoice, with 6-day average processing time and 3% error rate.

SOLUTION APPROACH

Document AI pipeline using LayoutLM for structured extraction from PDFs and scanned documents. Automated matching with PO system. Exception routing for human review.

PROJECT DETAILS

Domain	Finance / Operations
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · LayoutLM · HuggingFace · SAP AP API · Azure Form Recognizer · PostgreSQL

EXPECTED OUTCOMES

Cost per invoice reduced from $8 to $0.80. Processing time from 6 days to 4 hours. Error rate reduced to 0.4%.

P048 › Automated Invoice Processing — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LayoutLM
- HuggingFace
- SAP AP API
- Azure Form Recognizer
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Cost per invoice reduced from $8 to $0.80. Processing time from 6 days to 4 hours. Error rate reduced to 0.4%.

3	10 weeks	Finance
Team Size	Duration	Domain

PROBLEM CONTEXT

AP team manually processing 15,000 invoices/month at $8 per invoice, with 6-day average processing time and 3% error rate.

P049

Data Governance & Catalog Platform

Data Engineering

PROBLEM STATEMENT

2,000+ data assets with no lineage documentation, causing 6-week average onboarding for new data consumers and recurring compliance issues.

SOLUTION APPROACH

Apache Atlas for data catalog with automated metadata harvesting from Snowflake, Databricks, and Airflow. Business glossary with stewardship workflow.

PROJECT DETAILS

Domain	Financial Services / Data Platform
Team Size	4 people
Duration	4 months
Type	Data Engineering

TECHNOLOGY STACK

Apache Atlas · Python · Snowflake · Databricks · Apache Airflow · React

EXPECTED OUTCOMES

100% metadata coverage for critical assets. Onboarding time reduced from 6 weeks to 3 days. Zero compliance findings related to lineage.

P049 › Data Governance & Catalog Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Apache Atlas
- Python
- Snowflake
- Databricks
- Apache Airflow
- React

KEY METRICS & BUSINESS IMPACT

100% metadata coverage for critical assets. Onboarding time reduced from 6 weeks to 3 days. Zero compliance findings related to lineage.

4	4	Financial Services
Team Size	Duration	Domain

PROBLEM CONTEXT

2,000+ data assets with no lineage documentation, causing 6-week average onboarding for new data consumers and recurring compliance issues.

P050

Customer Lifetime Value Prediction

ML/AI

PROBLEM STATEMENT

Marketing budget allocated equally across all customers, wasting $3.2M annually on low-value segments with high acquisition cost.

SOLUTION APPROACH

BG/NBD + Gamma-Gamma model for transaction-based CLV. Segmentation into value tiers. Integration with marketing automation for budget allocation rules.

PROJECT DETAILS

Domain	E-Commerce / Subscription
Team Size	2 people
Duration	6 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · Lifetimes library · Scikit-learn · Snowflake · dbt · Braze API

EXPECTED OUTCOMES

Marketing ROI improved 34%. LTV prediction MAPE of 12% at 12-month horizon. $1.8M reallocation to high-value segments.

P050 › Customer Lifetime Value Prediction — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Lifetimes library
- Scikit-learn
- Snowflake
- dbt
- Braze API

KEY METRICS & BUSINESS IMPACT

Marketing ROI improved 34%. LTV prediction MAPE of 12% at 12-month horizon. $1.8M reallocation to high-value segments.

2	6 weeks	E-Commerce
Team Size	Duration	Domain

PROBLEM CONTEXT

Marketing budget allocated equally across all customers, wasting $3.2M annually on low-value segments with high acquisition cost.

P051

AI-Powered Meeting Summarization Tool

NLP / GenAI

PROBLEM STATEMENT

Employees spending 4+ hours/week writing meeting notes and follow-ups, with 40% of action items lost within 24 hours.

SOLUTION APPROACH

Whisper ASR for transcription + GPT-4 for structured summary extraction (decisions, actions, owners). Integrated with Zoom, MS Teams, and calendar systems.

PROJECT DETAILS

Domain	Enterprise / Productivity
Team Size	2 people
Duration	5 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI Whisper · GPT-4 · Python · Zoom SDK · MS Teams API · FastAPI

EXPECTED OUTCOMES

Meeting note creation time reduced from 45 min to 2 min. Action item follow-through rate improved from 60% to 91%.

P051 › AI-Powered Meeting Summarization Tool — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI Whisper
- GPT-4
- Python
- Zoom SDK
- MS Teams API
- FastAPI

KEY METRICS & BUSINESS IMPACT

Meeting note creation time reduced from 45 min to 2 min. Action item follow-through rate improved from 60% to 91%.

2	5 weeks	Enterprise
Team Size	Duration	Domain

PROBLEM CONTEXT

Employees spending 4+ hours/week writing meeting notes and follow-ups, with 40% of action items lost within 24 hours.

P052

Carbon Emissions Monitoring Platform

Data Engineering

PROBLEM STATEMENT

Manufacturing company unable to meet ESG reporting requirements due to fragmented Scope 1, 2, and 3 emissions data across 30 facilities.

SOLUTION APPROACH

Data integration pipeline from IoT sensors, utility APIs, and supply chain systems. Automated Scope 1/2/3 calculation per GHG Protocol. CDP/GRI reporting templates.

PROJECT DETAILS

Domain	Energy / Sustainability
Team Size	4 people
Duration	4 months
Type	Data Engineering

TECHNOLOGY STACK

Python · Apache Airflow · Snowflake · dbt · Power BI · IoT Hub

EXPECTED OUTCOMES

Full ESG reporting compliance achieved. Carbon intensity reduced 12% through identified optimization opportunities. CDP B rating achieved.

P052 › Carbon Emissions Monitoring Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Apache Airflow
- Snowflake
- dbt
- Power BI
- IoT Hub

KEY METRICS & BUSINESS IMPACT

Full ESG reporting compliance achieved. Carbon intensity reduced 12% through identified optimization opportunities. CDP B rating achieved.

4	4	Energy
Team Size	Duration	Domain

PROBLEM CONTEXT

Manufacturing company unable to meet ESG reporting requirements due to fragmented Scope 1, 2, and 3 emissions data across 30 facilities.

P053

NER-Based Clinical Trial Matching

NLP / GenAI

PROBLEM STATEMENT

Manual patient-to-trial matching taking 3–4 weeks per patient with only 12% enrollment rate on eligible candidates.

SOLUTION APPROACH

Clinical NLP pipeline extracting eligibility criteria from ClinicalTrials.gov. NER model for patient record extraction. Automated matching with explainability report.

PROJECT DETAILS

Domain	Healthcare / Pharma
Team Size	3 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · spaCy · MedSpaCy · BERT · PostgreSQL · FastAPI

EXPECTED OUTCOMES

Matching time reduced from 4 weeks to 2 days. Trial enrollment improved from 12% to 38%. Used across 5 hospital networks.

P053 › NER-Based Clinical Trial Matching — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- spaCy
- MedSpaCy
- BERT
- PostgreSQL
- FastAPI

KEY METRICS & BUSINESS IMPACT

Matching time reduced from 4 weeks to 2 days. Trial enrollment improved from 12% to 38%. Used across 5 hospital networks.

3	3	Healthcare
Team Size	Duration	Domain

PROBLEM CONTEXT

Manual patient-to-trial matching taking 3–4 weeks per patient with only 12% enrollment rate on eligible candidates.

P054

Multimodal Retail Analytics

ML/AI

PROBLEM STATEMENT

Brick-and-mortar retailer with no in-store behavior analytics, unable to optimize layout, staffing, or product placement.

SOLUTION APPROACH

Computer vision system using store cameras for footfall heatmaps, dwell time, and conversion funnel. Privacy-preserving (no facial recognition). Dashboard for store ops.

PROJECT DETAILS

Domain	Retail
Team Size	5 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · OpenCV · YOLOv8 · Apache Kafka · InfluxDB · Grafana

EXPECTED OUTCOMES

Store layout optimized based on heatmaps — 19% uplift in high-margin product sales. Staffing efficiency improved 15%.

P054 › Multimodal Retail Analytics — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- OpenCV
- YOLOv8
- Apache Kafka
- InfluxDB
- Grafana

KEY METRICS & BUSINESS IMPACT

Store layout optimized based on heatmaps — 19% uplift in high-margin product sales. Staffing efficiency improved 15%.

5	4	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

Brick-and-mortar retailer with no in-store behavior analytics, unable to optimize layout, staffing, or product placement.

P055

ML Pipeline Orchestration Platform

Data Engineering

PROBLEM STATEMENT

Data science team managing 50+ ML models in production with no centralized monitoring, causing silent model degradation and alert fatigue.

SOLUTION APPROACH

Built MLOps platform on Kubernetes with MLflow for experiment tracking, Evidently AI for drift detection, and custom alerting. Model registry with approval workflows.

PROJECT DETAILS

Domain	Technology / MLOps
Team Size	4 people
Duration	3 months
Type	Data Engineering

TECHNOLOGY STACK

Kubernetes · MLflow · Evidently AI · Python · Prometheus · Grafana

EXPECTED OUTCOMES

100% of production models monitored. 5 drift events caught proactively (vs. 0 previously). 30% reduction in model maintenance overhead.

P055 › ML Pipeline Orchestration Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Kubernetes
- MLflow
- Evidently AI
- Python
- Prometheus
- Grafana

KEY METRICS & BUSINESS IMPACT

100% of production models monitored. 5 drift events caught proactively (vs. 0 previously). 30% reduction in model maintenance overhead.

4	3	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

Data science team managing 50+ ML models in production with no centralized monitoring, causing silent model degradation and alert fatigue.

P056

Legal Case Outcome Prediction

NLP / GenAI

PROBLEM STATEMENT

Law firm's case assessment process taking 15+ hours per case with high subjectivity and 35% variance between senior partners.

SOLUTION APPROACH

NLP pipeline on 50K historical case documents. Fine-tuned Legal-BERT for outcome prediction with SHAP explanations. Calibrated probability scores for settlement strategy.

PROJECT DETAILS

Domain	Legal / Government
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · Legal-BERT · HuggingFace · SHAP · FastAPI · PostgreSQL

EXPECTED OUTCOMES

Case assessment time reduced from 15 hours to 2 hours. Prediction AUC 0.82. Settlement recommendation accuracy 73%.

P056 › Legal Case Outcome Prediction — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Legal-BERT
- HuggingFace
- SHAP
- FastAPI
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Case assessment time reduced from 15 hours to 2 hours. Prediction AUC 0.82. Settlement recommendation accuracy 73%.

3	10 weeks	Legal
Team Size	Duration	Domain

PROBLEM CONTEXT

Law firm's case assessment process taking 15+ hours per case with high subjectivity and 35% variance between senior partners.

P057

Automated A/B Testing Platform

Data Engineering

PROBLEM STATEMENT

Engineering team running 3–4 A/B tests simultaneously with manual p-value calculation and no guardrail metrics, causing $500K in regressions.

SOLUTION APPROACH

Self-serve experimentation platform with Bayesian sequential testing, automatic sample size calculation, and guardrail metric monitoring. Integrated with product analytics.

PROJECT DETAILS

Domain	Technology / Product
Team Size	3 people
Duration	3 months
Type	Data Engineering

TECHNOLOGY STACK

Python · FastAPI · PostgreSQL · React · Snowflake · Airflow

EXPECTED OUTCOMES

Test velocity increased from 4 to 25 concurrent experiments. Regressions eliminated by guardrail system. $2.1M in experiment-driven improvements.

P057 › Automated A/B Testing Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- FastAPI
- PostgreSQL
- React
- Snowflake
- Airflow

KEY METRICS & BUSINESS IMPACT

Test velocity increased from 4 to 25 concurrent experiments. Regressions eliminated by guardrail system. $2.1M in experiment-driven improvements.

3	3	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

Engineering team running 3–4 A/B tests simultaneously with manual p-value calculation and no guardrail metrics, causing $500K in regressions.

P058

Intelligent Procurement Analytics

Data Engineering

PROBLEM STATEMENT

Procurement team negotiating vendor contracts without spend analytics, paying 12–18% above benchmark prices across $200M in annual spend.

SOLUTION APPROACH

Spend analytics pipeline consolidating 8 ERP systems. Price benchmarking using external market data. Supplier risk scoring with news sentiment analysis.

PROJECT DETAILS

Domain	Manufacturing / Procurement
Team Size	3 people
Duration	3 months
Type	Data Engineering

TECHNOLOGY STACK

Python · Snowflake · dbt · Airflow · Power BI · News API

EXPECTED OUTCOMES

Identified $24M in savings opportunities. Achieved 8.4% average price reduction in renegotiations. Supplier risk visibility for top 200 vendors.

P058 › Intelligent Procurement Analytics — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Snowflake
- dbt
- Airflow
- Power BI
- News API

KEY METRICS & BUSINESS IMPACT

Identified $24M in savings opportunities. Achieved 8.4% average price reduction in renegotiations. Supplier risk visibility for top 200 vendors.

3	3	Manufacturing
Team Size	Duration	Domain

PROBLEM CONTEXT

Procurement team negotiating vendor contracts without spend analytics, paying 12–18% above benchmark prices across $200M in annual spend.

P059

Autonomous Crop Disease Detection

ML/AI

PROBLEM STATEMENT

Smallholder farmers in developing regions losing 30–40% of crop yield to preventable diseases due to no access to expert agronomists.

SOLUTION APPROACH

Mobile-first app with offline CNN model (MobileNetV3) detecting 26 crop diseases from photos. Edge deployment for areas with no connectivity. Hindi/Swahili multilingual UI.

PROJECT DETAILS

Domain	Agriculture / GovTech
Team Size	4 people
Duration	5 months
Type	ML/AI

TECHNOLOGY STACK

Python · TensorFlow Lite · MobileNetV3 · React Native · Firebase · AWS

EXPECTED OUTCOMES

Deployed to 120K+ farmers across 3 countries. Disease detection accuracy 91%. Average yield loss reduced from 35% to 12%.

P059 › Autonomous Crop Disease Detection — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- TensorFlow Lite
- MobileNetV3
- React Native
- Firebase
- AWS

KEY METRICS & BUSINESS IMPACT

Deployed to 120K+ farmers across 3 countries. Disease detection accuracy 91%. Average yield loss reduced from 35% to 12%.

4	5	Agriculture
Team Size	Duration	Domain

PROBLEM CONTEXT

Smallholder farmers in developing regions losing 30–40% of crop yield to preventable diseases due to no access to expert agronomists.

P060

Revenue Operations Data Platform

Data Engineering

PROBLEM STATEMENT

Revenue leaders making decisions from 3 disconnected dashboards (CRM, billing, marketing) with 3-day data lag and 15% discrepancy in reported MRR.

SOLUTION APPROACH

Unified RevOps data model in Snowflake combining Salesforce, Stripe, HubSpot, and Mixpanel. Single source of truth for ARR, churn, NRR, and pipeline metrics.

PROJECT DETAILS

Domain	B2B SaaS / RevOps
Team Size	3 people
Duration	10 weeks
Type	Data Engineering

TECHNOLOGY STACK

Snowflake · dbt · Fivetran · Salesforce · Stripe API · Looker

EXPECTED OUTCOMES

Single MRR source of truth with zero reconciliation issues. Real-time pipeline visibility. Reduced board reporting prep from 3 days to 2 hours.

P060 › Revenue Operations Data Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Snowflake
- dbt
- Fivetran
- Salesforce
- Stripe API
- Looker

KEY METRICS & BUSINESS IMPACT

Single MRR source of truth with zero reconciliation issues. Real-time pipeline visibility. Reduced board reporting prep from 3 days to 2 hours.

3	10 weeks	B2B SaaS
Team Size	Duration	Domain

PROBLEM CONTEXT

Revenue leaders making decisions from 3 disconnected dashboards (CRM, billing, marketing) with 3-day data lag and 15% discrepancy in reported MRR.

P061

Text-to-SQL Analytics Assistant

NLP / GenAI

PROBLEM STATEMENT

Business analysts blocked on data team queue (5-day wait) for SQL queries, slowing time-to-insight across 12 business units.

SOLUTION APPROACH

LLM-based text-to-SQL system using schema-aware prompting. Query validation layer to prevent dangerous mutations. Self-correcting mechanism for failed queries.

PROJECT DETAILS

Domain	Enterprise / Analytics
Team Size	2 people
Duration	6 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4 · Python · Snowflake · FastAPI · React · LangChain

EXPECTED OUTCOMES

80% of analytical queries now self-served. Data team queue reduced by 70%. Business analyst productivity +4 hours/week.

P061 › Text-to-SQL Analytics Assistant — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4
- Python
- Snowflake
- FastAPI
- React
- LangChain

KEY METRICS & BUSINESS IMPACT

80% of analytical queries now self-served. Data team queue reduced by 70%. Business analyst productivity +4 hours/week.

2	6 weeks	Enterprise
Team Size	Duration	Domain

PROBLEM CONTEXT

Business analysts blocked on data team queue (5-day wait) for SQL queries, slowing time-to-insight across 12 business units.

P062

Fleet Telematics & Driver Scoring Platform

ML/AI

PROBLEM STATEMENT

Fleet insurer pricing all drivers identically despite 5x variance in risk, leading to adverse selection and 22% loss ratio.

SOLUTION APPROACH

Telematics pipeline ingesting GPS, accelerometer, and braking data. Driver risk scoring with explainable features. Usage-based insurance pricing API.

PROJECT DETAILS

Domain	Logistics / Insurance
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · Apache Kafka · InfluxDB · XGBoost · FastAPI · PostgreSQL

EXPECTED OUTCOMES

Driver risk prediction AUC 0.84. Loss ratio improved from 22% to 14%. Enabled 15% premium discount for low-risk drivers.

P062 › Fleet Telematics & Driver Scoring Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Apache Kafka
- InfluxDB
- XGBoost
- FastAPI
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Driver risk prediction AUC 0.84. Loss ratio improved from 22% to 14%. Enabled 15% premium discount for low-risk drivers.

4	4	Logistics
Team Size	Duration	Domain

PROBLEM CONTEXT

Fleet insurer pricing all drivers identically despite 5x variance in risk, leading to adverse selection and 22% loss ratio.

P063

AI-Assisted Drug Interaction Checker

NLP / GenAI

PROBLEM STATEMENT

Pharmacists manually checking drug interactions for 200+ combinations per shift, with 2.3% missed interaction rate causing adverse events.

SOLUTION APPROACH

Knowledge graph of 15K drug interactions from DrugBank + FDA data. LLM-augmented explanation for clinical context. Integrated with hospital pharmacy system.

PROJECT DETAILS

Domain	Pharma / Healthcare
Team Size	3 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · Neo4j · OpenAI API · FastAPI · React · PostgreSQL

EXPECTED OUTCOMES

Missed interaction rate reduced to 0.1%. Alert fatigue reduced by 60% through severity scoring. Deployed across 12 hospital pharmacies.

P063 › AI-Assisted Drug Interaction Checker — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Neo4j
- OpenAI API
- FastAPI
- React
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Missed interaction rate reduced to 0.1%. Alert fatigue reduced by 60% through severity scoring. Deployed across 12 hospital pharmacies.

3	3	Pharma
Team Size	Duration	Domain

PROBLEM CONTEXT

Pharmacists manually checking drug interactions for 200+ combinations per shift, with 2.3% missed interaction rate causing adverse events.

P064

Personalized News Feed Ranking

ML/AI

PROBLEM STATEMENT

News platform with 12% daily active user retention vs. industry 28%, due to non-personalized content feed surfacing irrelevant articles.

SOLUTION APPROACH

Two-tower neural network for user-article matching. Thompson Sampling exploration for new content. Real-time feature store on Redis for sub-50ms recommendations.

PROJECT DETAILS

Domain	Media / Publishing
Team Size	5 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · TensorFlow · Redis · Apache Kafka · BigQuery · Kubernetes

EXPECTED OUTCOMES

DAU retention improved from 12% to 21%. Session length increased 34%. Article click-through improved from 3.1% to 7.8%.

P064 › Personalized News Feed Ranking — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- TensorFlow
- Redis
- Apache Kafka
- BigQuery
- Kubernetes

KEY METRICS & BUSINESS IMPACT

DAU retention improved from 12% to 21%. Session length increased 34%. Article click-through improved from 3.1% to 7.8%.

5	4	Media
Team Size	Duration	Domain

PROBLEM CONTEXT

News platform with 12% daily active user retention vs. industry 28%, due to non-personalized content feed surfacing irrelevant articles.

P065

Automated Tax Compliance Engine

NLP / GenAI

PROBLEM STATEMENT

Tax compliance team spending 2,000+ hours/year on indirect tax calculations across 40 jurisdictions, with $800K in penalty exposure.

SOLUTION APPROACH

Rules engine with NLP-based tax code parsing. Automated VAT/GST calculation integrated with billing system. Audit trail generation and jurisdiction-specific filing.

PROJECT DETAILS

Domain	FinTech / RegTech
Team Size	3 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · spaCy · FastAPI · PostgreSQL · Stripe Tax API · React

EXPECTED OUTCOMES

Tax calculation time reduced from 2,000 hours to 40 hours/year. Zero compliance penalties for 18 months post-launch.

P065 › Automated Tax Compliance Engine — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- spaCy
- FastAPI
- PostgreSQL
- Stripe Tax API
- React

KEY METRICS & BUSINESS IMPACT

Tax calculation time reduced from 2,000 hours to 40 hours/year. Zero compliance penalties for 18 months post-launch.

3	3	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Tax compliance team spending 2,000+ hours/year on indirect tax calculations across 40 jurisdictions, with $800K in penalty exposure.

P066

Seismic Event Classification System

ML/AI

PROBLEM STATEMENT

Geophysics team manually classifying 5,000+ seismic events/day, with 6-hour classification lag delaying drilling decisions.

SOLUTION APPROACH

1D CNN trained on seismic waveform data for event classification (earthquake vs. noise vs. blast). AutoML hyperparameter optimization. Real-time streaming inference.

PROJECT DETAILS

Domain	Energy / O&G
Team Size	3 people
Duration	10 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · TensorFlow · 1D CNN · Apache Kafka · InfluxDB · Grafana

EXPECTED OUTCOMES

Classification accuracy 96.4%. Processing time from 6 hours to real-time. Enabled $8M drilling decision made 2 days faster.

P066 › Seismic Event Classification System — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- TensorFlow
- 1D CNN
- Apache Kafka
- InfluxDB
- Grafana

KEY METRICS & BUSINESS IMPACT

Classification accuracy 96.4%. Processing time from 6 hours to real-time. Enabled $8M drilling decision made 2 days faster.

3	10 weeks	Energy
Team Size	Duration	Domain

PROBLEM CONTEXT

Geophysics team manually classifying 5,000+ seismic events/day, with 6-hour classification lag delaying drilling decisions.

P067

Customer Segmentation & Personalization Engine

ML/AI

PROBLEM STATEMENT

Marketing using 4 static customer segments built 3 years ago, resulting in 1.2% email CTR vs. industry 4.5% benchmark.

SOLUTION APPROACH

Unsupervised clustering (UMAP + HDBSCAN) on behavioral and transactional data yielding 22 micro-segments. Automated journey orchestration per segment.

PROJECT DETAILS

Domain	Retail / Banking
Team Size	3 people
Duration	2 months
Type	ML/AI

TECHNOLOGY STACK

Python · UMAP · HDBSCAN · Scikit-learn · Snowflake · Braze

EXPECTED OUTCOMES

Email CTR improved from 1.2% to 4.1%. Campaign revenue uplift 28%. Segment refresh now automated weekly vs. annually.

P067 › Customer Segmentation & Personalization Engine — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- UMAP
- HDBSCAN
- Scikit-learn
- Snowflake
- Braze

KEY METRICS & BUSINESS IMPACT

Email CTR improved from 1.2% to 4.1%. Campaign revenue uplift 28%. Segment refresh now automated weekly vs. annually.

3	2	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

Marketing using 4 static customer segments built 3 years ago, resulting in 1.2% email CTR vs. industry 4.5% benchmark.

P068

Serverless Data Lakehouse on AWS

Data Engineering

PROBLEM STATEMENT

Streaming platform running $800K/month in always-on EMR clusters for analytical workloads that run 6 hours/day.

SOLUTION APPROACH

Migrated to serverless stack: S3 + Iceberg tables + Athena for query + Step Functions for orchestration. EMR Serverless for Spark jobs. Cost tagging via resource groups.

PROJECT DETAILS

Domain	Media / Streaming
Team Size	4 people
Duration	3 months
Type	Data Engineering

TECHNOLOGY STACK

AWS S3 · Apache Iceberg · Amazon Athena · AWS Step Functions · EMR Serverless · Terraform

EXPECTED OUTCOMES

Infrastructure cost reduced from $800K to $190K/month (76% reduction). Query performance improved 2.3x. Zero idle compute waste.

P068 › Serverless Data Lakehouse on AWS — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- AWS S3
- Apache Iceberg
- Amazon Athena
- AWS Step Functions
- EMR Serverless
- Terraform

KEY METRICS & BUSINESS IMPACT

Infrastructure cost reduced from $800K to $190K/month (76% reduction). Query performance improved 2.3x. Zero idle compute waste.

4	3	Media
Team Size	Duration	Domain

PROBLEM CONTEXT

Streaming platform running $800K/month in always-on EMR clusters for analytical workloads that run 6 hours/day.

P069

Synthetic Data Generation for Model Training

ML/AI

PROBLEM STATEMENT

ML team unable to share training data across team boundaries due to PII regulations, blocking model development for 3 high-priority projects.

SOLUTION APPROACH

CTGAN-based tabular synthetic data generator with statistical fidelity tests. Privacy guarantees via differential privacy. Automated fitness-for-purpose evaluation.

PROJECT DETAILS

Domain	FinTech / Healthcare
Team Size	3 people
Duration	10 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · CTGAN · SDV Library · TensorFlow Privacy · Great Expectations · FastAPI

EXPECTED OUTCOMES

3 blocked ML projects unblocked within 2 weeks of synthetic data availability. Statistical fidelity >95% vs. real data.

P069 › Synthetic Data Generation for Model Training — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- CTGAN
- SDV Library
- TensorFlow Privacy
- Great Expectations
- FastAPI

KEY METRICS & BUSINESS IMPACT

3 blocked ML projects unblocked within 2 weeks of synthetic data availability. Statistical fidelity >95% vs. real data.

3	10 weeks	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

ML team unable to share training data across team boundaries due to PII regulations, blocking model development for 3 high-priority projects.

P070

Intelligent Incident Management System

NLP / GenAI

PROBLEM STATEMENT

IT team triaging 2,000+ monthly incidents manually, with 35-minute average MTTA and frequent misrouting to wrong teams.

SOLUTION APPROACH

NLP classification for automatic incident routing (92% accuracy). Similar incident retrieval via FAISS for suggested resolutions. Auto-escalation based on severity prediction.

PROJECT DETAILS

Domain	Technology / ITSM
Team Size	3 people
Duration	2 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · DistilBERT · FAISS · FastAPI · ServiceNow API · Elasticsearch

EXPECTED OUTCOMES

MTTA reduced from 35 to 8 minutes. Misrouting reduced by 89%. 34% of incidents auto-resolved via suggested fixes.

P070 › Intelligent Incident Management System — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- DistilBERT
- FAISS
- FastAPI
- ServiceNow API
- Elasticsearch

KEY METRICS & BUSINESS IMPACT

MTTA reduced from 35 to 8 minutes. Misrouting reduced by 89%. 34% of incidents auto-resolved via suggested fixes.

3	2	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

IT team triaging 2,000+ monthly incidents manually, with 35-minute average MTTA and frequent misrouting to wrong teams.

P071

Precision Agriculture Water Management

ML/AI

PROBLEM STATEMENT

Irrigation system over-watering crops by 35% due to fixed schedules, wasting $400K/year in water and reducing yield quality.

SOLUTION APPROACH

IoT soil moisture sensor network with LSTM model for irrigation scheduling. Integration with weather API for forecast-based planning. Mobile app for farmer alerts.

PROJECT DETAILS

Domain	Agriculture / IoT
Team Size	3 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · LSTM · MQTT · AWS IoT · InfluxDB · React Native

EXPECTED OUTCOMES

Water usage reduced by 31%. Crop yield quality improved 14%. ROI payback period 8 months.

P071 › Precision Agriculture Water Management — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LSTM
- MQTT
- AWS IoT
- InfluxDB
- React Native

KEY METRICS & BUSINESS IMPACT

Water usage reduced by 31%. Crop yield quality improved 14%. ROI payback period 8 months.

3	3	Agriculture
Team Size	Duration	Domain

PROBLEM CONTEXT

Irrigation system over-watering crops by 35% due to fixed schedules, wasting $400K/year in water and reducing yield quality.

P072

Real-Time Risk Monitoring for Trading Desk

Data Engineering

PROBLEM STATEMENT

Risk desk running end-of-day batch VaR calculations with no intraday risk visibility, exposing firm to undetected intraday limit breaches.

SOLUTION APPROACH

Streaming risk calculation engine on Apache Flink processing 50K trades/second. Real-time P&L attribution and Greeks calculation. Breach alerting with trader dashboards.

PROJECT DETAILS

Domain	Capital Markets
Team Size	6 people
Duration	6 months
Type	Data Engineering

TECHNOLOGY STACK

Apache Flink · Python · Redis · Apache Kafka · Grafana · PostgreSQL

EXPECTED OUTCOMES

Real-time VaR calculation with 200ms latency. 3 intraday limit breaches caught in first month (none before). Regulatory compliance achieved.

P072 › Real-Time Risk Monitoring for Trading Desk — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Apache Flink
- Python
- Redis
- Apache Kafka
- Grafana
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Real-time VaR calculation with 200ms latency. 3 intraday limit breaches caught in first month (none before). Regulatory compliance achieved.

6	6	Capital Markets
Team Size	Duration	Domain

PROBLEM CONTEXT

Risk desk running end-of-day batch VaR calculations with no intraday risk visibility, exposing firm to undetected intraday limit breaches.

P073

Conversational Commerce Platform

NLP / GenAI

PROBLEM STATEMENT

WhatsApp-heavy customer base (60% prefer WhatsApp for support) unable to complete purchases in-channel, causing 35% drop-off.

SOLUTION APPROACH

WhatsApp Business API integration with product catalog, cart management, and payment flow. LLM-powered natural language order handling with human fallback.

PROJECT DETAILS

Domain	E-Commerce / Retail
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · WhatsApp Business API · LangChain · FastAPI · Stripe API · PostgreSQL

EXPECTED OUTCOMES

In-WhatsApp purchase conversion: 14%. $2.1M monthly GMV through WhatsApp channel. Cart abandonment reduced 45%.

P073 › Conversational Commerce Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- WhatsApp Business API
- LangChain
- FastAPI
- Stripe API
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

In-WhatsApp purchase conversion: 14%. $2.1M monthly GMV through WhatsApp channel. Cart abandonment reduced 45%.

3	10 weeks	E-Commerce
Team Size	Duration	Domain

PROBLEM CONTEXT

WhatsApp-heavy customer base (60% prefer WhatsApp for support) unable to complete purchases in-channel, causing 35% drop-off.

P074

AI Governance & Model Risk Framework

Data Engineering

PROBLEM STATEMENT

Bank deploying 30+ ML models without standardized risk assessment, creating regulatory exposure ahead of SR 11-7 audit.

SOLUTION APPROACH

Model governance platform with automated bias testing (Fairlearn), explainability reports (SHAP/LIME), performance monitoring, and approval workflow.

PROJECT DETAILS

Domain	Financial Services / Risk
Team Size	4 people
Duration	4 months
Type	Data Engineering

TECHNOLOGY STACK

Python · Fairlearn · SHAP · LIME · MLflow · React · PostgreSQL

EXPECTED OUTCOMES

30 models documented and risk-rated. SR 11-7 audit passed with no findings. Model deployment SLA standardized to 2 weeks.

P074 › AI Governance & Model Risk Framework — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Fairlearn
- SHAP
- LIME
- MLflow
- React
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

30 models documented and risk-rated. SR 11-7 audit passed with no findings. Model deployment SLA standardized to 2 weeks.

4	4	Financial Services
Team Size	Duration	Domain

PROBLEM CONTEXT

Bank deploying 30+ ML models without standardized risk assessment, creating regulatory exposure ahead of SR 11-7 audit.

P075

Multilingual Customer Support Automation

NLP / GenAI

PROBLEM STATEMENT

Customer support handling queries in 18 languages with 12-hour SLA, requiring expensive human agents in each region.

SOLUTION APPROACH

Multilingual RAG system using mT5 embeddings over localized FAQ knowledge base. LLM response generation with tone calibration per culture. 8-language live agent handoff.

PROJECT DETAILS

Domain	Travel / Hospitality
Team Size	4 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · mT5 · HuggingFace · LangChain · Zendesk API · FastAPI

EXPECTED OUTCOMES

18-language support with <5 minute SLA. 72% automation rate. Human agent cost reduced by $1.8M annually.

P075 › Multilingual Customer Support Automation — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- mT5
- HuggingFace
- LangChain
- Zendesk API
- FastAPI

KEY METRICS & BUSINESS IMPACT

18-language support with <5 minute SLA. 72% automation rate. Human agent cost reduced by $1.8M annually.

4	3	Travel
Team Size	Duration	Domain

PROBLEM CONTEXT

Customer support handling queries in 18 languages with 12-hour SLA, requiring expensive human agents in each region.

P076

Anomaly Detection for Smart Grid

ML/AI

PROBLEM STATEMENT

Utility detecting energy theft and meter tampering via manual inspections, catching only 15% of incidents at $2M annual loss.

SOLUTION APPROACH

Isolation Forest + LSTM autoencoder for consumption pattern anomaly detection. Alert prioritization using risk scoring. Integration with field service dispatch system.

PROJECT DETAILS

Domain	Energy / Utilities
Team Size	3 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · Scikit-learn · LSTM · Apache Kafka · PostgreSQL · Power BI

EXPECTED OUTCOMES

Energy theft detection improved from 15% to 71%. $1.4M annual loss recovered. 90% reduction in false positive field dispatches.

P076 › Anomaly Detection for Smart Grid — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Scikit-learn
- LSTM
- Apache Kafka
- PostgreSQL
- Power BI

KEY METRICS & BUSINESS IMPACT

Energy theft detection improved from 15% to 71%. $1.4M annual loss recovered. 90% reduction in false positive field dispatches.

3	3	Energy
Team Size	Duration	Domain

PROBLEM CONTEXT

Utility detecting energy theft and meter tampering via manual inspections, catching only 15% of incidents at $2M annual loss.

P077

B2B Lead Scoring & Intent Prediction

ML/AI

PROBLEM STATEMENT

Sales team working all inbound leads equally, with 80% of deals closed from 20% of leads — but no way to identify which 20%.

SOLUTION APPROACH

Gradient boosting model on firmographic, behavioral, and intent data (G2, Bombora). Real-time scoring via webhook into Salesforce. SDR prioritization dashboard.

PROJECT DETAILS

Domain	B2B SaaS / Marketing
Team Size	2 people
Duration	6 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · XGBoost · Salesforce API · Bombora API · FastAPI · Snowflake

EXPECTED OUTCOMES

Lead-to-opportunity conversion improved 2.4x for top-scored leads. SDR capacity freed by 30%. $3.8M incremental pipeline in Q1.

P077 › B2B Lead Scoring & Intent Prediction — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- XGBoost
- Salesforce API
- Bombora API
- FastAPI
- Snowflake

KEY METRICS & BUSINESS IMPACT

Lead-to-opportunity conversion improved 2.4x for top-scored leads. SDR capacity freed by 30%. $3.8M incremental pipeline in Q1.

2	6 weeks	B2B SaaS
Team Size	Duration	Domain

PROBLEM CONTEXT

Sales team working all inbound leads equally, with 80% of deals closed from 20% of leads — but no way to identify which 20%.

P078

Document Redaction & PII Masking Pipeline

NLP / GenAI

PROBLEM STATEMENT

Discovery team manually redacting PII from 50,000 documents/month at $12/doc, with 4.2% missed PII rate creating legal liability.

SOLUTION APPROACH

NLP-based PII detection using ensemble of spaCy NER + Presidio. Confidence-based auto-redaction with human review for low-confidence entities. Audit-ready redaction log.

PROJECT DETAILS

Domain	Legal / Compliance
Team Size	2 people
Duration	8 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · spaCy · Microsoft Presidio · FastAPI · Azure Blob · React

EXPECTED OUTCOMES

Redaction cost reduced from $12 to $0.45/doc. Missed PII rate reduced to 0.08%. 180K documents processed in first 3 months.

P078 › Document Redaction & PII Masking Pipeline — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- spaCy
- Microsoft Presidio
- FastAPI
- Azure Blob
- React

KEY METRICS & BUSINESS IMPACT

Redaction cost reduced from $12 to $0.45/doc. Missed PII rate reduced to 0.08%. 180K documents processed in first 3 months.

2	8 weeks	Legal
Team Size	Duration	Domain

PROBLEM CONTEXT

Discovery team manually redacting PII from 50,000 documents/month at $12/doc, with 4.2% missed PII rate creating legal liability.

P079

HR Attrition Risk Analytics

ML/AI

PROBLEM STATEMENT

High attrition (22% annually) among senior engineers with no early warning capability, resulting in $1.2M average replacement cost per senior hire.

SOLUTION APPROACH

Survival analysis model using Kaplan-Meier + Cox proportional hazards on HR data. Individual risk scores with SHAP-based retention intervention recommendations.

PROJECT DETAILS

Domain	HR / Enterprise
Team Size	2 people
Duration	5 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · Lifelines · SHAP · Snowflake · dbt · Tableau

EXPECTED OUTCOMES

At-risk engineers identified 3–4 months in advance. Attrition reduced from 22% to 14% with targeted interventions. $4.8M annual savings.

P079 › HR Attrition Risk Analytics — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Lifelines
- SHAP
- Snowflake
- dbt
- Tableau

KEY METRICS & BUSINESS IMPACT

At-risk engineers identified 3–4 months in advance. Attrition reduced from 22% to 14% with targeted interventions. $4.8M annual savings.

2	5 weeks	HR
Team Size	Duration	Domain

PROBLEM CONTEXT

High attrition (22% annually) among senior engineers with no early warning capability, resulting in $1.2M average replacement cost per senior hire.

P080

AI-Powered News Credibility Checker

NLP / GenAI

PROBLEM STATEMENT

News aggregator platform failing to filter misinformation, resulting in 8% of content flagged by users as potentially false.

SOLUTION APPROACH

Claim verification pipeline: NLP claim extraction → web search → cross-reference against trusted sources → credibility score via ensemble classifier.

PROJECT DETAILS

Domain	Media / Trust & Safety
Team Size	3 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · OpenAI API · Google Search API · FastAPI · PostgreSQL · React

EXPECTED OUTCOMES

Misinformation rate reduced from 8% to 1.2%. 94% precision on credibility scoring. 0 PR incidents in 9 months post-launch.

P080 › AI-Powered News Credibility Checker — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- OpenAI API
- Google Search API
- FastAPI
- PostgreSQL
- React

KEY METRICS & BUSINESS IMPACT

Misinformation rate reduced from 8% to 1.2%. 94% precision on credibility scoring. 0 PR incidents in 9 months post-launch.

3	3	Media
Team Size	Duration	Domain

PROBLEM CONTEXT

News aggregator platform failing to filter misinformation, resulting in 8% of content flagged by users as potentially false.

P081

Predictive Asset Inspection Scheduler

ML/AI

PROBLEM STATEMENT

City managing 8,000 road and bridge assets with fixed 2-year inspection cycles, missing critical deterioration events between cycles.

SOLUTION APPROACH

Condition-based inspection scheduling using deterioration models. Satellite imagery change detection for early warning. Risk-prioritized maintenance scheduling.

PROJECT DETAILS

Domain	Infrastructure / Government
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · Scikit-learn · Google Earth Engine · PostGIS · FastAPI · React

EXPECTED OUTCOMES

Critical deterioration detected 14 months earlier on average. Maintenance budget efficiency improved 28%. 0 infrastructure incidents in pilot area.

P081 › Predictive Asset Inspection Scheduler — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Scikit-learn
- Google Earth Engine
- PostGIS
- FastAPI
- React

KEY METRICS & BUSINESS IMPACT

Critical deterioration detected 14 months earlier on average. Maintenance budget efficiency improved 28%. 0 infrastructure incidents in pilot area.

4	4	Infrastructure
Team Size	Duration	Domain

PROBLEM CONTEXT

City managing 8,000 road and bridge assets with fixed 2-year inspection cycles, missing critical deterioration events between cycles.

P082

Real-Time Translation for Global Support

NLP / GenAI

PROBLEM STATEMENT

Global support platform serving 60 countries using human translators with $1.2M annual cost and 2-hour average translation lag.

SOLUTION APPROACH

Neural machine translation pipeline using Helsinki-NLP models fine-tuned on domain-specific terminology. Real-time translation in support ticketing system with confidence scores.

PROJECT DETAILS

Domain	Enterprise / Globalization
Team Size	2 people
Duration	8 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · HuggingFace Helsinki-NLP · FastAPI · Redis · Zendesk API · Docker

EXPECTED OUTCOMES

Translation cost reduced from $1.2M to $80K/year. Translation latency from 2 hours to 800ms. BLEU score 34 (vs. 27 for generic MT).

P082 › Real-Time Translation for Global Support — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- HuggingFace Helsinki-NLP
- FastAPI
- Redis
- Zendesk API
- Docker

KEY METRICS & BUSINESS IMPACT

Translation cost reduced from $1.2M to $80K/year. Translation latency from 2 hours to 800ms. BLEU score 34 (vs. 27 for generic MT).

2	8 weeks	Enterprise
Team Size	Duration	Domain

PROBLEM CONTEXT

Global support platform serving 60 countries using human translators with $1.2M annual cost and 2-hour average translation lag.

P083

Micro-Lending Credit Scoring for Emerging Markets

ML/AI

PROBLEM STATEMENT

MFI lending to 500K borrowers using traditional bureau scores that exclude 70% of target population (unbanked individuals).

SOLUTION APPROACH

Alternative credit scoring using mobile money transactions, airtime top-up patterns, and psychometric quiz. Fairness-constrained model to avoid demographic proxy features.

PROJECT DETAILS

Domain	FinTech / Development Finance
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · LightGBM · Fairlearn · Mobile Money API · PostgreSQL · FastAPI

EXPECTED OUTCOMES

Credit access extended to 150K previously excluded borrowers. Default rate maintained at 2.1%. Model gini coefficient 0.62.

P083 › Micro-Lending Credit Scoring for Emerging Markets — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- LightGBM
- Fairlearn
- Mobile Money API
- PostgreSQL
- FastAPI

KEY METRICS & BUSINESS IMPACT

Credit access extended to 150K previously excluded borrowers. Default rate maintained at 2.1%. Model gini coefficient 0.62.

4	4	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

MFI lending to 500K borrowers using traditional bureau scores that exclude 70% of target population (unbanked individuals).

P084

Graph-Based Recommendation for OTT

ML/AI

PROBLEM STATEMENT

OTT platform with 8% monthly churn driven by poor content recommendations surfacing already-watched content for 28% of sessions.

SOLUTION APPROACH

Graph neural network recommendation using watch history, genre, and collaborative signals. Diversity constraint in ranking to avoid filter bubble effect.

PROJECT DETAILS

Domain	Media / Streaming
Team Size	5 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · PyTorch Geometric · Redis · Apache Kafka · Kubernetes · BigQuery

EXPECTED OUTCOMES

Monthly churn reduced from 8% to 5.3%. Content diversity score +41%. Watch time increased 23%.

P084 › Graph-Based Recommendation for OTT — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch Geometric
- Redis
- Apache Kafka
- Kubernetes
- BigQuery

KEY METRICS & BUSINESS IMPACT

Monthly churn reduced from 8% to 5.3%. Content diversity score +41%. Watch time increased 23%.

5	4	Media
Team Size	Duration	Domain

PROBLEM CONTEXT

OTT platform with 8% monthly churn driven by poor content recommendations surfacing already-watched content for 28% of sessions.

P085

Field Service Optimization Platform

ML/AI

PROBLEM STATEMENT

Field service scheduling manual, resulting in 35% travel time waste and $4.2M annual cost from inefficient technician routing.

SOLUTION APPROACH

MILP-based scheduling optimization with dynamic rescheduling on job completion/cancellation. Skills matching and parts availability integration. Mobile app for technicians.

PROJECT DETAILS

Domain	Utilities / Field Operations
Team Size	5 people
Duration	5 months
Type	ML/AI

TECHNOLOGY STACK

Python · PuLP (MILP) · Google OR-Tools · FastAPI · React Native · PostgreSQL

EXPECTED OUTCOMES

Travel time reduced by 27%. Service jobs completed per technician per day improved from 4.2 to 6.1. $2.8M annual savings.

P085 › Field Service Optimization Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PuLP (MILP)
- Google OR-Tools
- FastAPI
- React Native
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Travel time reduced by 27%. Service jobs completed per technician per day improved from 4.2 to 6.1. $2.8M annual savings.

5	5	Utilities
Team Size	Duration	Domain

PROBLEM CONTEXT

Field service scheduling manual, resulting in 35% travel time waste and $4.2M annual cost from inefficient technician routing.

P086

NLP Pipeline for Earnings Call Analysis

NLP / GenAI

PROBLEM STATEMENT

Equity research team manually transcribing and analyzing 200+ earnings calls/quarter, taking 3 days per company for sentiment and topic extraction.

SOLUTION APPROACH

Whisper ASR for transcription + FinBERT for financial sentiment. Topic modeling via BERTopic. Automated summary and tone-change detection vs. prior quarter.

PROJECT DETAILS

Domain	Finance / Investment
Team Size	2 people
Duration	6 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · Whisper · FinBERT · BERTopic · FastAPI · PostgreSQL

EXPECTED OUTCOMES

Earnings call analysis time reduced from 3 days to 45 minutes. Tone change signals correlated with 3-day stock moves at 0.62 correlation.

P086 › NLP Pipeline for Earnings Call Analysis — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Whisper
- FinBERT
- BERTopic
- FastAPI
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Earnings call analysis time reduced from 3 days to 45 minutes. Tone change signals correlated with 3-day stock moves at 0.62 correlation.

2	6 weeks	Finance
Team Size	Duration	Domain

PROBLEM CONTEXT

Equity research team manually transcribing and analyzing 200+ earnings calls/quarter, taking 3 days per company for sentiment and topic extraction.

P087

Smart Home Energy Optimization

ML/AI

PROBLEM STATEMENT

Smart home platform users seeing 0% energy savings despite having sensors, due to no intelligent automation or usage insights.

SOLUTION APPROACH

Disaggregation model identifying appliance-level consumption from whole-home smart meter data. RL agent for HVAC scheduling. In-app recommendations with savings calculator.

PROJECT DETAILS

Domain	IoT / Energy
Team Size	3 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · NILMTK · Stable Baselines3 · MQTT · InfluxDB · React Native

EXPECTED OUTCOMES

Average household energy savings of 18%. App engagement increased 3x. 92% of users report increased awareness of consumption patterns.

P087 › Smart Home Energy Optimization — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- NILMTK
- Stable Baselines3
- MQTT
- InfluxDB
- React Native

KEY METRICS & BUSINESS IMPACT

Average household energy savings of 18%. App engagement increased 3x. 92% of users report increased awareness of consumption patterns.

3	3	IoT
Team Size	Duration	Domain

PROBLEM CONTEXT

Smart home platform users seeing 0% energy savings despite having sensors, due to no intelligent automation or usage insights.

P088

AI-Driven Market Research Synthesis

NLP / GenAI

PROBLEM STATEMENT

Research team spending 6 weeks synthesizing 500+ pages of market research per engagement, with subjective and inconsistent outputs.

SOLUTION APPROACH

RAG pipeline over research documents + web sources. LLM-based synthesis with structured output (key findings, market sizing, competitive landscape). Citation-grounded.

PROJECT DETAILS

Domain	Consulting / Market Research
Team Size	2 people
Duration	5 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4 · LangChain · FAISS · Python · Streamlit · PDF Parser

EXPECTED OUTCOMES

Research synthesis time reduced from 6 weeks to 3 days. Client satisfaction improved. 2 additional engagements won citing speed advantage.

P088 › AI-Driven Market Research Synthesis — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4
- LangChain
- FAISS
- Python
- Streamlit
- PDF Parser

KEY METRICS & BUSINESS IMPACT

Research synthesis time reduced from 6 weeks to 3 days. Client satisfaction improved. 2 additional engagements won citing speed advantage.

2	5 weeks	Consulting
Team Size	Duration	Domain

PROBLEM CONTEXT

Research team spending 6 weeks synthesizing 500+ pages of market research per engagement, with subjective and inconsistent outputs.

P089

Microgrid Load Balancing with RL

ML/AI

PROBLEM STATEMENT

Solar microgrid operator experiencing 22% energy waste due to suboptimal battery charge/discharge scheduling against time-of-use pricing.

SOLUTION APPROACH

Deep RL agent (PPO) trained on historical load, solar generation, and pricing data. Deployed on edge controller with cloud retraining. Safety constraints for battery health.

PROJECT DETAILS

Domain	Energy / CleanTech
Team Size	3 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · Stable Baselines3 (PPO) · TensorFlow · MQTT · InfluxDB · Grafana

EXPECTED OUTCOMES

Energy waste reduced by 19%. Electricity cost reduced 24% through ToU arbitrage. Battery lifecycle extended by estimated 18%.

P089 › Microgrid Load Balancing with RL — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Stable Baselines3 (PPO)
- TensorFlow
- MQTT
- InfluxDB
- Grafana

KEY METRICS & BUSINESS IMPACT

Energy waste reduced by 19%. Electricity cost reduced 24% through ToU arbitrage. Battery lifecycle extended by estimated 18%.

3	4	Energy
Team Size	Duration	Domain

PROBLEM CONTEXT

Solar microgrid operator experiencing 22% energy waste due to suboptimal battery charge/discharge scheduling against time-of-use pricing.

P090

Multi-Source Data Integration Platform

Data Engineering

PROBLEM STATEMENT

Hospital network with 15 EMR systems preventing population health analytics and value-based care reporting.

SOLUTION APPROACH

FHIR R4-compliant data integration layer normalizing records from 15 source systems. Master patient index with probabilistic matching. HL7 v2 to FHIR converter.

PROJECT DETAILS

Domain	Healthcare / Data Platform
Team Size	6 people
Duration	7 months
Type	Data Engineering

TECHNOLOGY STACK

Python · FHIR R4 · Apache Kafka · PostgreSQL · dbt · Azure

EXPECTED OUTCOMES

Unified patient records for 2.3M patients. Population health programs launched across 3 care gaps. HEDIS measure compliance improved 15%.

P090 › Multi-Source Data Integration Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- FHIR R4
- Apache Kafka
- PostgreSQL
- dbt
- Azure

KEY METRICS & BUSINESS IMPACT

Unified patient records for 2.3M patients. Population health programs launched across 3 care gaps. HEDIS measure compliance improved 15%.

6	7	Healthcare
Team Size	Duration	Domain

PROBLEM CONTEXT

Hospital network with 15 EMR systems preventing population health analytics and value-based care reporting.

P091

Toxic Content Detection for Social Platform

NLP / GenAI

PROBLEM STATEMENT

Social platform with 2M DAU seeing 4% content toxicity rate causing user churn, with moderation team overwhelmed by 50K+ daily reports.

SOLUTION APPROACH

Multilabel toxicity classifier (Perspective API + custom fine-tuned model). Context-aware detection for sarcasm and dog-whistles. Graduated response automation.

PROJECT DETAILS

Domain	Trust & Safety / Social
Team Size	3 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · Perspective API · HuggingFace · FastAPI · PostgreSQL · Redis

EXPECTED OUTCOMES

Toxicity rate reduced from 4% to 0.7%. Moderation team queue reduced 80%. User retention improved 9%.

P091 › Toxic Content Detection for Social Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Perspective API
- HuggingFace
- FastAPI
- PostgreSQL
- Redis

KEY METRICS & BUSINESS IMPACT

Toxicity rate reduced from 4% to 0.7%. Moderation team queue reduced 80%. User retention improved 9%.

3	3	Trust & Safety
Team Size	Duration	Domain

PROBLEM CONTEXT

Social platform with 2M DAU seeing 4% content toxicity rate causing user churn, with moderation team overwhelmed by 50K+ daily reports.

P092

Supply Chain Risk Intelligence Platform

NLP / GenAI

PROBLEM STATEMENT

Procurement team blind to geopolitical and ESG risks in supplier base until crises occur, causing two major supply disruptions in 12 months ($45M impact).

SOLUTION APPROACH

Automated news and regulatory filing ingestion with NLP risk classification. Supplier risk scoring across 7 dimensions. Early warning system with 30-day risk horizon.

PROJECT DETAILS

Domain	Manufacturing / Risk
Team Size	4 people
Duration	3 months
Type	NLP / GenAI

TECHNOLOGY STACK

Python · HuggingFace · News API · Snowflake · Airflow · Power BI

EXPECTED OUTCOMES

2 supply risks identified 6 weeks before materialization in first year. Risk coverage extended to 450 Tier-1 and Tier-2 suppliers.

P092 › Supply Chain Risk Intelligence Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- HuggingFace
- News API
- Snowflake
- Airflow
- Power BI

KEY METRICS & BUSINESS IMPACT

2 supply risks identified 6 weeks before materialization in first year. Risk coverage extended to 450 Tier-1 and Tier-2 suppliers.

4	3	Manufacturing
Team Size	Duration	Domain

PROBLEM CONTEXT

Procurement team blind to geopolitical and ESG risks in supplier base until crises occur, causing two major supply disruptions in 12 months ($45M impact).

P093

Automated Sports Highlight Generation

ML/AI

PROBLEM STATEMENT

Sports broadcaster manually editing 3-hour match recordings to create 5-minute highlight reels, taking 4 hours per match with 8 matches per weekend.

SOLUTION APPROACH

Computer vision + audio analysis pipeline detecting key moments (goals, fouls, crowd reactions). LLM-based commentary generation. Automated video stitching.

PROJECT DETAILS

Domain	Media / Sports
Team Size	4 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · OpenCV · YOLOv8 · Whisper · OpenAI GPT-4 · FFmpeg

EXPECTED OUTCOMES

Highlight generation time reduced from 4 hours to 18 minutes. Social media engagement on automated highlights 91% of manual quality.

P093 › Automated Sports Highlight Generation — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- OpenCV
- YOLOv8
- Whisper
- OpenAI GPT-4
- FFmpeg

KEY METRICS & BUSINESS IMPACT

Highlight generation time reduced from 4 hours to 18 minutes. Social media engagement on automated highlights 91% of manual quality.

4	4	Media
Team Size	Duration	Domain

PROBLEM CONTEXT

Sports broadcaster manually editing 3-hour match recordings to create 5-minute highlight reels, taking 4 hours per match with 8 matches per weekend.

P094

Predictive Spare Parts Inventory

ML/AI

PROBLEM STATEMENT

Maintenance team experiencing 18% stockout rate on critical spare parts causing $2.8M in extended downtime annually.

SOLUTION APPROACH

Intermittent demand forecasting using Croston's method + bootstrapping simulation. Min-max policy optimization per part per warehouse. SAP integration for auto-PO.

PROJECT DETAILS

Domain	Manufacturing / MRO
Team Size	2 people
Duration	6 weeks
Type	ML/AI

TECHNOLOGY STACK

Python · Statsmodels · Scipy · SAP MM API · PostgreSQL · Power BI

EXPECTED OUTCOMES

Stockout rate reduced from 18% to 3%. Inventory carrying cost reduced 22%. $2.1M in downtime costs avoided in Year 1.

P094 › Predictive Spare Parts Inventory — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Statsmodels
- Scipy
- SAP MM API
- PostgreSQL
- Power BI

KEY METRICS & BUSINESS IMPACT

Stockout rate reduced from 18% to 3%. Inventory carrying cost reduced 22%. $2.1M in downtime costs avoided in Year 1.

2	6 weeks	Manufacturing
Team Size	Duration	Domain

PROBLEM CONTEXT

Maintenance team experiencing 18% stockout rate on critical spare parts causing $2.8M in extended downtime annually.

P095

Geospatial Market Expansion Analytics

ML/AI

PROBLEM STATEMENT

Retail chain opening new stores based on intuition, with 30% of stores in last 3 years failing to hit Year 1 revenue targets.

SOLUTION APPROACH

Site selection model using geospatial features: footfall (Safegraph), competitor density, demographic fit, cannibalization radius. Interactive suitability map.

PROJECT DETAILS

Domain	Retail / Strategy
Team Size	3 people
Duration	3 months
Type	ML/AI

TECHNOLOGY STACK

Python · GeoPandas · Safegraph API · XGBoost · Folium · PostGIS

EXPECTED OUTCOMES

Model adopted for 12 new store decisions. Year 1 revenue target hit rate improved from 70% to 92%. Avoided 2 estimated failures.

P095 › Geospatial Market Expansion Analytics — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- GeoPandas
- Safegraph API
- XGBoost
- Folium
- PostGIS

KEY METRICS & BUSINESS IMPACT

Model adopted for 12 new store decisions. Year 1 revenue target hit rate improved from 70% to 92%. Avoided 2 estimated failures.

3	3	Retail
Team Size	Duration	Domain

PROBLEM CONTEXT

Retail chain opening new stores based on intuition, with 30% of stores in last 3 years failing to hit Year 1 revenue targets.

P096

Real-Time Bid Optimization for Ad Platform

ML/AI

PROBLEM STATEMENT

Programmatic advertising bidding with fixed CPM, resulting in 40% of ad spend on low-ROI inventory and $6M in wasted budget.

SOLUTION APPROACH

Contextual bandit algorithm for real-time bid pricing. 50ms inference on feature store. CVR prediction model integrated into bidder. Budget pacing controller.

PROJECT DETAILS

Domain	AdTech / Marketing
Team Size	5 people
Duration	4 months
Type	ML/AI

TECHNOLOGY STACK

Python · Vowpal Wabbit · Redis · FastAPI · Kafka · Spark

EXPECTED OUTCOMES

Ad spend efficiency improved 38%. CPA reduced from $42 to $26. $3.2M annual savings on same budget with higher conversions.

P096 › Real-Time Bid Optimization for Ad Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- Vowpal Wabbit
- Redis
- FastAPI
- Kafka
- Spark

KEY METRICS & BUSINESS IMPACT

Ad spend efficiency improved 38%. CPA reduced from $42 to $26. $3.2M annual savings on same budget with higher conversions.

5	4	AdTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Programmatic advertising bidding with fixed CPM, resulting in 40% of ad spend on low-ROI inventory and $6M in wasted budget.

P097

Automated ESG Scoring for Investment Portfolio

NLP / GenAI

PROBLEM STATEMENT

ESG investment fund manually scoring 400+ portfolio companies on ESG metrics, taking 3 months per annual review cycle.

SOLUTION APPROACH

NLP pipeline extracting ESG indicators from sustainability reports, news, and regulatory filings. Automated scoring with evidence-based justification and controversy flag.

PROJECT DETAILS

Domain	Finance / ESG
Team Size	3 people
Duration	10 weeks
Type	NLP / GenAI

TECHNOLOGY STACK

Python · HuggingFace · SEC EDGAR API · News API · FastAPI · Snowflake

EXPECTED OUTCOMES

Annual scoring cycle reduced from 3 months to 2 weeks. Coverage expanded from 400 to 1,200 companies. Used in $2.4B fund investment committee.

P097 › Automated ESG Scoring for Investment Portfolio — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- HuggingFace
- SEC EDGAR API
- News API
- FastAPI
- Snowflake

KEY METRICS & BUSINESS IMPACT

Annual scoring cycle reduced from 3 months to 2 weeks. Coverage expanded from 400 to 1,200 companies. Used in $2.4B fund investment committee.

3	10 weeks	Finance
Team Size	Duration	Domain

PROBLEM CONTEXT

ESG investment fund manually scoring 400+ portfolio companies on ESG metrics, taking 3 months per annual review cycle.

P098

3D Point Cloud Processing for Autonomous Vehicles

ML/AI

PROBLEM STATEMENT

AV development team processing LiDAR point cloud data at 2 FPS — far below 10 FPS minimum needed for safe real-time object detection.

SOLUTION APPROACH

PointNet++ model with custom CUDA kernels for GPU acceleration. Voxelization optimization reducing memory footprint by 60%. Object detection reaching 12 FPS.

PROJECT DETAILS

Domain	Automotive / Robotics
Team Size	5 people
Duration	5 months
Type	ML/AI

TECHNOLOGY STACK

Python · PyTorch · CUDA · PointNet++ · ROS2 · Docker

EXPECTED OUTCOMES

Processing speed improved from 2 FPS to 12 FPS. Object detection mAP: 0.81. Safety milestone for on-road testing unlocked.

P098 › 3D Point Cloud Processing for Autonomous Vehicles — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Python
- PyTorch
- CUDA
- PointNet++
- ROS2
- Docker

KEY METRICS & BUSINESS IMPACT

Processing speed improved from 2 FPS to 12 FPS. Object detection mAP: 0.81. Safety milestone for on-road testing unlocked.

5	5	Automotive
Team Size	Duration	Domain

PROBLEM CONTEXT

AV development team processing LiDAR point cloud data at 2 FPS — far below 10 FPS minimum needed for safe real-time object detection.

P099

Personalized Financial Planning Chatbot

NLP / GenAI

PROBLEM STATEMENT

Wealth management firm unable to serve mass-affluent segment (<$500K AUM) profitably with human advisors, missing $180M addressable market.

SOLUTION APPROACH

Conversational AI for financial goal setting, risk profiling, and portfolio recommendation. SEC-compliant advice guardrails. Human advisor escalation for complex scenarios.

PROJECT DETAILS

Domain	FinTech / Wealth Management
Team Size	4 people
Duration	5 months
Type	NLP / GenAI

TECHNOLOGY STACK

OpenAI GPT-4 · LangChain · Python · FastAPI · React · PostgreSQL

EXPECTED OUTCOMES

Mass-affluent segment launched. 8,500 users onboarded in 3 months. AUM under robo-advisory: $42M. Compliance audit passed.

P099 › Personalized Financial Planning Chatbot — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- OpenAI GPT-4
- LangChain
- Python
- FastAPI
- React
- PostgreSQL

KEY METRICS & BUSINESS IMPACT

Mass-affluent segment launched. 8,500 users onboarded in 3 months. AUM under robo-advisory: $42M. Compliance audit passed.

4	5	FinTech
Team Size	Duration	Domain

PROBLEM CONTEXT

Wealth management firm unable to serve mass-affluent segment (<$500K AUM) profitably with human advisors, missing $180M addressable market.

P100

Unified MLOps & Feature Platform

Data Engineering

PROBLEM STATEMENT

6 ML teams maintaining separate feature pipelines with 70% feature redundancy and 3-week average time-to-production for new models.

SOLUTION APPROACH

Centralized feature store on Feast + Redis/Snowflake. Shared feature registry with reuse metrics. Standardized model packaging with BentoML. CI/CD for ML pipelines.

PROJECT DETAILS

Domain	Technology / Platform Engineering
Team Size	5 people
Duration	6 months
Type	Data Engineering

TECHNOLOGY STACK

Feast · Redis · Snowflake · BentoML · Kubernetes · GitHub Actions

EXPECTED OUTCOMES

Feature reuse rate improved from 30% to 74%. Model deployment time reduced from 3 weeks to 2 days. Infrastructure cost reduced 40%.

P100 › Unified MLOps & Feature Platform — Technical Deep Dive

FULL TECHNOLOGY BREAKDOWN

- Feast
- Redis
- Snowflake
- BentoML
- Kubernetes
- GitHub Actions

KEY METRICS & BUSINESS IMPACT

Feature reuse rate improved from 30% to 74%. Model deployment time reduced from 3 weeks to 2 days. Infrastructure cost reduced 40%.

5	6	Technology
Team Size	Duration	Domain

PROBLEM CONTEXT

6 ML teams maintaining separate feature pipelines with 70% feature redundancy and 3-week average time-to-production for new models.